

TABLE 6

Ref.	a. core radius (mm)	b. radius of winding structure (mm)	c. segment skip region (mm)	d. height variable region (mm)	e. stack number uniform region (mm)	f. segment region (mm)	g. uniform region stack number	c/(b - a) (%)	d/f (%)	e/f (%)	h. ratio of segment skip region	i. ratio of height variable region	j. ratio of height uniform region
embodiment 6-1	4	22	3	2	11	15	13	17%	13%	73%	6%	7%	83%
embodiment 6-2	4	22	3	3	10	15	16	17%	20%	67%	6%	11%	80%
embodiment 6-3	4	22	3	4	9	15	18	17%	27%	60%	6%	15%	75%
embodiment 6-4	4	22	3	5	8	15	21	17%	33%	53%	6%	21%	69%
embodiment 6-5	4	22	3	6	7	15	24	17%	40%	47%	6%	25%	65%
embodiment 6-6	4	22	3	7	6	15	26	17%	47%	40%	6%	32%	59%
embodiment 7-1	2	22	3	2	13	17	13	15%	12%	76%	6%	7%	83%
embodiment 7-2	2	22	3	3	12	17	16	15%	18%	71%	6%	11%	80%
embodiment 7-3	2	22	3	4	11	17	18	15%	24%	65%	6%	15%	75%
embodiment 7-4	2	22	3	5	10	17	21	15%	29%	59%	6%	21%	69%
embodiment 7-5	2	22	3	6	9	17	24	15%	35%	53%	6%	25%	65%
embodiment 7-6	2	22	3	7	8	17	26	15%	41%	47%	6%	32%	59%

Seeing FIG. 17a and the embodiments 6-6 and 7-6 of Table 6, the minimum height (h_1) and the maximum height (h_N) of the segments in the height variable region (2) of the segments are the same as 3 mm and 10 mm, respectively. However, in the embodiment 6-6, the radius of the core is larger by 2 mm than that in the embodiment 7-6. Therefore, in the embodiment 6-6, the stack number uniform region (e) and the segment region (f) are smaller by 2 mm than those in the embodiment 7-6, and the stack number of segments is the same in the stack number uniform region. This result comes from the difference in the radius of the core. From the results shown in Table 6, when the width of the height variable region (2) of the segments is the same, it may be understood that, as the radius (a) of the core is smaller, the ratio (d/f) of the height variable region (2) decreases, but the ratio (e/f) of the stack number uniform region increases.

Seeing Table 6, it may be found that the stack number of segments is 13 to 26, the ratio (d/f) of the height variable region (2) of the segments is 12% to 47%, and the ratio (e/f) of the length of the stack number uniform region is 40% to 76%. In addition, the ratio (c/(b-a)) of the segment skip region (1) to the radius (b-a) of the electrode assembly excluding the core is 15% to 17%. In addition, the ratio of the length of the electrode area corresponding to the segment skip region (1) to the entire length of the electrode is 6%, the ratio of the length of the electrode area corresponding to the height variable region (2) to the entire length of the electrode is 7% to 32%, and the ratio of the length of the electrode area corresponding to the height uniform region (3) to the entire length of the electrode is 59% to 83%.

For cylindrical batteries with form factors of 1865 and 2170, the radius of the electrode assembly is approximately 9 mm to 10 mm. Therefore, different from the embodiments, the length of the segment region (f) in the radial direction is not secured at the level of 15 mm to 17 mm, and at the same time the length of the stack number uniform region (e) where the stack number of segments is 10 or more cannot be secured at the level of 6 mm to 13 mm, while securing the

length of the segment skip region (1) at the level of about 3 mm. This is because, in the conventional cylindrical battery, when the radius of the core is designed to be 2 mm to 4 mm, which is the same as the embodiments, the radial region in which segments can be disposed is substantially only 5 mm to 8 mm. In addition, in the conventional cylindrical battery, the length of the electrode in the winding direction is about 600 mm to 980 mm. This short length of the electrode is only about 15% to 24% of the length of the electrode (positive electrode 3948 mm, negative electrode 4045 mm) in the embodiments. Therefore, the numerical ranges for the parameters h, i, and j cannot be easily derived from design specifications of the conventional cylindrical batteries.

Comprehensively considering the data in Tables 4 to 6, the stack number of segments may be 11 to 26 in the stack number uniform region of the segments. In addition, the ratio (d/f) of the height variable region (2) of the segments may be 6% to 47%. Also, the ratio (e/f) of the stack number uniform region may be 31% to 82%. In addition, the ratio (c/(b-a)) of the length of the segment skip region (1) to the radius of the electrode assembly excluding the core may be 15% to 35%. In addition, the ratio of the length of the electrode area corresponding to the segment skip region (1) to the entire length (length in the winding direction) of the electrode may be 6% to 20%. In addition, the ratio of the length of the electrode area corresponding to the height variable region (2) of the segments to the entire length of the electrode may be 3% to 32%. In addition, the ratio of the length of the electrode area corresponding to the height uniform region (3) of the segments to the entire length of the electrode may be 59% to 87%.

Meanwhile, the parameters described in Tables 4 to 6 are be varied according to design factors including the radius (a) of the core; the radius of the electrode assembly B; the minimum height (h_1) and the maximum height (hp) in the height variable region (2) of the segments; the height change amount (Δh) of the segments per 1 mm increment of

the radius; the thickness of the positive electrode, the negative electrode and the separator, and the like.

Therefore, in the stack number uniform region of the segments, the segment stack number may be 10 to 35. The ratio (d/f) of the height variable region (2) of the segments may be 1% to 50%. Also, the ratio (e/f) of the stack number uniform region may be 30% to 85%. In addition, the ratio (c/(b-a)) of the length of the segment skip region (1) to the radius of the electrode assembly excluding the core may be 10% to 40%. In addition, the ratio of the length of the electrode area corresponding to the segment skip region (1) to the entire length (length in the winding direction) of the electrode may be 1% to 30%. In addition, the ratio of the length of the electrode area corresponding to the height variable region (2) of the segments to the entire length of the electrode may be 1% to 40%. In addition, the ratio of the length of the electrode area corresponding to the height uniform region (3) of the segments to the entire length of the electrode may be 50% to 90%. In the embodiments, the height index N of the maximum height (h_m) of the segments included in the height variable region (2) and the height uniform region (3) is 2 to 8. For example, seeing Table 4, the height index N for the embodiments 1-1 and 1-7 is 2 and 8, respectively. However, the height index N may vary depending on the height change amount (Δh) of the segment in the radial direction of the electrode assembly. When the radial length of the height variable region (2) is fixed, if the height change amount (Δh) of the segment decreases, the height index N increases accordingly, or vice versa. Preferably, the height index N may be further expanded to 2 to 20, optionally to 2 to 30.

In the bending surface region F formed at the upper portion and the lower portion of the electrode assembly, the stack number uniform region may be used as the welding target area of the current collector.

Preferably, the welding region of the current collector overlaps the stack number uniform region by at least 50% in the radial direction of the electrode assembly, and a higher overlapping ratio is more preferred.

Preferably, the rest of the area of the welding region of the current collector that does not overlap with the stack number uniform region may overlap with the stack number decrease region adjacent to the stack number uniform region in the radial direction.

More preferably, the rest of the area of the welding region of the current collector that does not overlap with the stack number uniform region may overlap with the area of the stack number decrease region in which the segment stack number is 10 or more.

If the current collector is welded to the area where the segment stack number is 10 or more, it is desirable in terms of the welding strength and prevention of damage to the separator or the active material layer during welding. In particular, it is useful when welding the current collector using a high-power laser with high transmission characteristics.

If the stack number uniform region where 10 or more of the segments are stacked and the current collector are welded with a laser, even if the output of the laser is increased to improve welding quality, the stack number uniform region absorbs most of the laser energy to form a welding bead, so it is possible to prevent the separator and the active material layer below the bending surface region F from being damaged by the laser.

In addition, since the segment stack number is 10 or more in the area where the laser is irradiated, welding beads are formed with sufficient volume and thickness. Therefore,

sufficient welding strength may be secured and the resistance of the welding interface may be reduced to a level suitable for rapid charging.

When welding the current collector, the output of the laser may be determined by the desired welding strength between the bending surface region F and the current collector. The welding strength increases in proportion to the stack number of segments. This is because the volume of the welding beads formed by the laser increases as the stack number increases. The welding beads are formed as the material of the current collector and the material of the segment are melted together. Therefore, when the volume of the welding bead is large, the current collector and the bending surface region are coupled stronger and the contact resistance of the welding interface is lowered.

Preferably, the welding strength may be 2 kgf/cm² or more, more preferably 4 kgf/cm² or more. The maximum welding strength may vary depending on the output of the laser welding equipment. Also, the welding strength may be preferably set to 8 kgf/cm² or less, more preferably 6 kgf/cm² or less. However, the present disclosure is not limited thereto.

When the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and the resistance of the welding interface may be reduced since the volume of the welding beads is sufficient.

The power of the laser to meet the welding strength condition differs depending on the laser equipment, and may be appropriately adjusted in the range of 250 W to 320 W or in the range of 40% to 100% of the laser maximum output provided by the equipment.

The welding strength may be defined as a tensile force (kgf/cm²) per unit area of the current collector when the current collector starts to separate from the bending surface region F. Specifically, after the current collector is completely welded, a tensile force may be applied to the current collector while gradually increasing the magnitude of the tensile force. When the tensile force exceeds a threshold value, the segment starts to separate from the welding interface. At this time, the value obtained by dividing the tensile force applied to the current collector by the area of the current collector corresponds to the welding strength.

In the bending surface region F, the segments are stacked in a plurality of layers, and according to the above embodiments, the stack number of segments may increase to 10 at minimum to 35 at maximum.

The thickness of the positive electrode current collector (foil) constituting the uncoated portion 43 is 10 μ m to 25 μ m, and the thickness of the negative electrode current collector (foil) constituting the uncoated portion 43 may be 5 μ m to 20 μ m. Therefore, the bending surface region F of the positive electrode may include an area where the total stack thickness of the segments is 100 μ m to 875 μ m. In addition, the bending surface region F of the negative electrode may include an area where the total layered thickness of the segments is 50 μ m to 700 μ m.

FIG. 17f is a top plan view of the electrode assembly showing the stack number uniform region b1 and the stack number decrease region b2 in the bending surface region F of the segments 61, 61' according to an embodiment of the present disclosure.

Referring to FIG. 17f, the area between two circles indicated by the thick solid line corresponds to the bending surface region F of the segments, and the area between two

circles indicated by the dashed-dotted line corresponds to the stack number uniform region b1 in which the stack number of the segments is 10 or more, and the outer area of the stack number uniform region b1 corresponds to the stack number decrease region b2.

In one example, when the current collector (P_c) is welded to the bending surface region F, a welding pattern (W_p) is generated on the surface of the current collector (P_c). The welding pattern (W_p) may have an array of line patterns or dot patterns. The welding pattern (W_p) corresponds to the welding region and may overlap by 50% or more with the stack number uniform region b1 of the segments along the radial direction. Therefore, a part of the welding pattern (W_p) may be included in the stack number uniform region b1, and the rest of the welding pattern (W_p) may be included in the stack number decrease region b2 outside the stack number uniform region b1. Of course, the entire welding pattern (W_p) may overlap with the stack number uniform region b1 in order to maximize the welding strength and lower the resistance of the welding region.

The area of the bending surface region F may be defined as the sum of the area of the stack number uniform region b1 and the area of the stack number decrease region b2 of the segment. Since the ratio (e/f) of the stack number uniform region b1 is 30% to 85%, preferably 31% to 82%, the ratio of the area of the stack number uniform region b1 to the area of the bending surface region F may be 9% ($30^2/100^2$) to 72% ($85^2/100^2$), preferably 10% ($31^2/100^2$) to 67% ($82^2/100^2$).

Preferably, the edge of the portion where the current collector (P_c) contacts the bending surface region F may cover the end of the segment 61, 61' bent toward the core C in the last winding turn of the height uniform region (3). In this case, since the welding pattern (W_p) is formed in a state where the segments 61, 61' are pressed by the current collector (P_c), the current collector (P_c) and the bending surface region F are strongly coupled. As a result, since the segments 61, 61' stacked in the winding axis direction come into close contact with each other, the resistance at the welding interface may be lowered and lifting of the segments 61, 61' may be prevented.

Meanwhile, the bending direction of the segments may be opposite to that described above. That is, the segments may be bent from the core toward the outer circumference. In this case, the pattern in which the heights of the segments change along the winding direction (X-axis direction) may be opposite to that of the embodiments (modifications) described above. For example, the heights of the segments may gradually decrease from the core toward the outer circumference. Also, the structure applied to the first portion B1 and the structure applied to the second portion B3 may be switched with each other. Preferably, the height change pattern may be designed such that the heights of the segments are gradually decreased from the core toward the outer circumference, but when the segment closest to the outer circumference of the electrode assembly is bent toward the outer circumference, the end of the segment does not protrude out of the outer circumference of the electrode assembly.

The electrode structure of the above embodiments (modifications) may be applied to at least one of the first electrode and the second electrode having different polarities included in the jelly-roll type electrode assembly or another type electrode assembly known in the art. In addition, when the electrode structure of the above embodiments (modifications) is applied to any one of the first electrode and the second electrode, the conventional electrode structure may

be applied to the other one. In addition, the electrode structures applied to the first electrode and the second electrode may not be identical but be different from each other.

For example, when the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, any one of the above embodiments (modifications) may be applied to the first electrode and the conventional electrode structure (see FIG. 1) may be applied to the second electrode.

As another example, when the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, any one of the above embodiments (modifications) may be selectively applied to the first electrode and any one of the above embodiments (modifications) may be selectively applied to the second electrode.

Hereinafter, the structure of the electrode assembly according to an embodiment of the present disclosure will be described in detail.

FIG. 18 is a cross-sectional view of a jelly-roll type electrode assembly 80 in which the electrode 40 according to the first embodiment is applied to a first electrode (positive electrode) and a second electrode (negative electrode), taken along the Y-axis direction (winding axis direction).

The electrode assembly 80 may be manufactured by the winding method described with reference to FIG. 2. For convenience of description, the protruding structure of the first uncoated portion 43a and the second uncoated portion 43b extending to the outside of the separator is shown in detail, and the winding structure of the first electrode, the second electrode, and the separator is not depicted. The first uncoated portion 43a upwardly protruding extends from the first electrode, and the second uncoated portion 43b downwardly protruding extends from the second electrode.

A pattern in which the heights of the first and second uncoated portions 43a, 43b change is schematically shown. That is, the height of the uncoated portion may change irregularly depending on the position where the section is cut. For example, if the side of the trapezoid segment 61, 61' or the cut groove 63 is cut, the height of the uncoated portion in the cross section is lower than the height (H) of the segment 61, 61'. Accordingly, it should be understood that the height of the uncoated portion shown in the cross-sectional drawings of the electrode assembly corresponds to the average of the heights (H in FIGS. 14b and 15b) of the uncoated portions included in each winding turn.

Referring to FIG. 18, the first uncoated portion 43a includes a first portion B1 adjacent to the core of the electrode assembly 80, a second portion B3 adjacent to the outer circumferential surface of the electrode assembly 80, and a third portion B2 interposed between the first portion B1 and the second portion B3.

The height (length in the Y-axis direction) of the second portion B3 is relatively smaller than the height of the third portion B2. Therefore, while the beading portion of the battery housing is being pressed near the second portion B3, it is possible to prevent a phenomenon in which an internal short circuit occurs as the beading portion and the second portion B3 contact each other.

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or the electrode structure in other embodiments (modifications).

The ends 81 of the first uncoated portion 43a and the second uncoated portion 43b may be bent in the radial direction of the electrode assembly 80, for example from the

outer circumference toward the core. At this time, the second portion B3 may not be substantially bent.

FIG. 19 is a cross-sectional view of a jelly-roll type electrode assembly 90 in which the electrode 45 of the second embodiment is applied to the first electrode (positive electrode) and the second electrode (negative electrode), taken along the Y-axis direction (winding axis direction).

Referring to FIG. 19, the first uncoated portion 43a of the first electrode includes a first portion B1 adjacent to the core of the electrode assembly 90, a second portion B3 adjacent to the outer circumferential surface of the electrode assembly 90, and a third portion B2 interposed between the first portion B1 and the second portion B3.

The height of the second portion B3 is relatively smaller than that of the third portion B2 and gradually or stepwise decreases from the core toward the outer circumference. Therefore, while the beading portion of the battery housing is being pressed near the second portion B3, it is possible to prevent a phenomenon in which an internal short circuit occurs as the beading portion and the second portion B3 contact each other.

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or an electrode structure of other embodiments (modifications).

The ends 91 of the first uncoated portion 43a and the second uncoated portion 43b may be bent in the radial direction of the electrode assembly 90, for example from the outer circumference toward the core. At this time, the outermost side 92 of the second portion B3 may not be substantially bent.

FIG. 20 is a cross-sectional view of a jelly-roll type electrode assembly 100 in which any one of the electrodes 50, 60, 70 of the third to fifth embodiments (modifications thereof) is applied to the first electrode (positive electrode) and the second electrode (negative electrode), taken along the Y-axis direction (winding axis direction).

Referring to FIG. 20, the uncoated portion 43a of the first electrode includes a first portion B1 adjacent to the core of the electrode assembly 100, a second portion B3 adjacent to the surface of the outer circumference of the electrode assembly 100, and a third portion B2 interposed between the first portion B1 and the second portion B3.

The height of the first portion B1 is relatively smaller than that of the third portion B2. Also, in the third portion B2, the bending length of the uncoated portion 43a located at the innermost side is equal to or smaller than the radial length (R) of the first portion B1. The bending length (H) corresponds to a distance from the point where the uncoated portion 43a is bent to a top end of the uncoated portion 43a. In a modification, the bending length H may be less than the sum of the radial length (R) of the first portion B1 and 10% of the radius of the core 102.

Therefore, even if the third portion B2 is bent, more than 90% of the diameter of the core 102 of the electrode assembly 100 is open to the outside. The core 102 is a cavity at the center of the electrode assembly 100. If the core 102 is not blocked, there is no difficulty in the electrolyte injection process and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the core 102, a welding process between the current collector at the negative electrode (or, the positive electrode) and the battery housing (or, the terminal) may be easily performed.

The height of the second portion B3 is relatively smaller than the height of the third portion B2. Therefore, while the beading portion of the battery housing is being pressed near

the second portion B3, it is possible to prevent a phenomenon in which an internal short circuit occurs as the beading portion and the second portion B3 contact each other.

In one modification, the height of the second portion B3 may decrease gradually or stepwise, unlike that shown in FIG. 20. Also, in FIG. 20, the height of the third portion B2 is the same in a part near the outer circumference, but the height of the third portion B2 may increase gradually or stepwise from the boundary between the first portion B1 and the third portion B2 toward the boundary between the third portion B2 and the second portion B3. When the third portion B2 is divided into a plurality of segments, a region in which the height of the uncoated portion 43a changes corresponds to the segment height variable region (②) in FIG. 17a).

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or an electrode structure of other embodiments (modifications).

The ends 101 of the first uncoated portion 43a and the second uncoated portion 43b may be bent in the radial direction of the electrode assembly 100, for example from the outer circumference toward the core. At this time, the first portion B1 and the second portion B3 are not substantially bent.

When the third portion B2 includes a plurality of segments, the bending stress may be alleviated to prevent the uncoated portions 43a near the bending point from being torn or abnormally deformed. In addition, when the width and/or height and/or separation pitch of the segments is adjusted according to the numerical range of the above embodiment, the segments are bent toward the core and overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region.

FIG. 21 is a cross-sectional view of an electrode assembly 110 according to still another embodiment of the present disclosure, taken along the Y-axis direction (winding axis direction).

Referring to FIG. 21, the electrode assembly 110 has substantially the same configuration as the electrode assembly 100 of FIG. 20, except that the height of the second portion B3 is substantially equal to the height of the outermost side of the third portion B2.

The second portion B3 may include a plurality of segments. The configuration of the plurality of segments is substantially the same as that of the fourth and fifth embodiments (modifications) of the electrode.

In the electrode assembly 110, the height of the first portion B1 is relatively smaller than the height of the third portion B2. Also, in the third portion B2, the bending length (H) of the uncoated portion located at the innermost side is equal to or smaller than the radial length (R) of the first portion B1. Preferably, the first portion B1 may be a segment skip region (①) in FIG. 17a) with no segment. In a modification, the bending length H may be less than the sum of the radial length (R) of the first portion B1 and 10% of the radius of the core 112.

Therefore, even if the third portion B2 is bent, 90% or more of the diameter of the core 112 of the electrode assembly 110 is open to the outside. If the core 112 is not blocked, there is no difficulty in the electrolyte injection process and the electrolyte injection efficiency is improved. In addition, the welding process between the current collector at the negative electrode (or, the positive electrode)

and the battery housing (or, the terminal) may be easily performed by inserting a welding jig through the core 112.

In one modification, the structure in which the height of the third portion B2 gradually or stepwise increases from the core toward the outer circumference may be extended to the second portion B3. In this case, the height of the uncoated portion 43a may gradually or stepwise increase from the boundary between the first portion B1 and the third portion B2 toward the surface of the outermost side of the electrode assembly 110.

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or an electrode structure of other embodiments (modifications).

The ends 111 of the first uncoated portion 43a and the second uncoated portion 43b may be bent in the radial direction of the electrode assembly 110, for example from the outer circumference toward the core. At this time, the first portion B1 is not substantially bent.

When the third portion B2 and the second portion B3 include a plurality of segments, the bending stress is relieved, so it is possible to prevent tearing or abnormal deformation of the uncoated portions 43a, 43b near the bending point. In addition, when the width and/or height and/or separation pitch of the segment is adjusted according to the numerical ranges of the above embodiment, the segments are bent toward the core and overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region.

FIG. 22 is a cross-sectional view showing the electrode assembly 120 according to still another embodiment of the present disclosure, taken along the Y-axis direction (winding axis direction).

Referring to FIG. 22, the electrode assembly 120 is substantially identical to the electrode assembly 100 of FIG. 20, except that the heights of the third portion B2 have a pattern of increasing gradually or stepwise and then decreasing. The radial region in which the heights of the third portion B2 change may be regarded as the height variable region (2) in FIG. 17a) of the segments. Even in this case, the height variable region of the segments may be designed so that the stack number uniform region in which the stack number of the segments is 10 or more appears in the desirable numerical range described above in the bending surface region F formed by bending the third portion B2.

The change in height of the third portion B2 may be implemented by adjusting the height of the step pattern (see FIG. 14c) or the segment (see FIG. 14d or 15a) included in the third portion B2.

In the electrode assembly 120, the height of the first portion B1 is relatively smaller than the height of the third portion B2. In addition, in the third portion B2, the bending length H of the uncoated portion located at the innermost side is equal to or smaller than the radial length R of the first portion B1. The region corresponding to the first portion B1 corresponds to the segment skip region (1) in FIG. 17a) with no segment. In one modification, the bending length H may be less than the sum of the radial length R of the first portion B1 and 10% of the radius of the core 122.

Therefore, even if the third portion B2 is bent toward the core, 90% or more of the diameter of the core 122 of the electrode assembly 120 is open to the outside. If the core 122 is not blocked, there is no difficulty in the electrolyte injection process, and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the

core 122, the welding process may be easily performed between the current collector of the negative electrode (or, positive electrode) and the battery housing (or, rivet terminal).

Also, the height of the second portion B3 is relatively smaller than the heights of the third portion B2, and preferably, the segment may not be formed in the second portion B3. Therefore, it is possible to prevent the phenomenon that the beading portion and the second portion B3 come into contact with each other to cause an internal short circuit while the beading portion of the battery housing is being pressed near the second portion B3. In one modification, the height of the second portion B3 may decrease gradually or stepwise toward the outer circumference.

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or an electrode structure of other embodiments (modifications).

The ends 121 of the first uncoated portion 43a and the second uncoated portion 43b may be bent from the outer circumference of the electrode assembly 120 toward the core. At this time, the first portion B1 and the second portion B3 are substantially not bent.

When the third portion B2 includes a plurality of segments, the bending stress is alleviated to prevent the uncoated portions 43a, 43b from being torn or abnormally deformed. In addition, when the width and/or height and/or separation pitch of the segments is adjusted according to the numerical range of the above embodiment, the segments are bent toward the core and overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region.

FIG. 23 is a cross-sectional view showing the electrode assembly 130 according to still another embodiment of the present disclosure, taken along the Y-axis direction (winding axis direction).

Referring to FIG. 23, the electrode assembly 130 is substantially identical to the electrode assembly 120 of FIG. 22, except that the height of the second portion B3 has a pattern of decreasing gradually or stepwise from the boundary point of the second portion B3 and the third portion B2 toward the outermost surface of the electrode assembly 130.

The change in height of the second portion B3 may be implemented by extending the step pattern (see FIG. 14c) included in the third portion B2 to the second portion B3 and at the same time gradually or stepwise decreasing the height of the pattern toward the outer circumference. Also, in another modification, the change in height of the second portion B3 may be implemented by extending the segment structure of the third portion B2 to the second portion B3 and at the same time gradually or stepwise decreasing the height of the segment toward the outer circumference.

In the electrode assembly 120, the height of the first portion B1 is relatively smaller than the height of the third portion B2. In addition, in the third portion B2, the bending length H of the uncoated portion located at the innermost side is equal to or smaller than the radial length R of the first portion B1. The first portion B1 corresponds to the segment skip region (1) in FIG. 17a) with no segment. In one modification, the bending length H may be less than the sum of the radial length R of the first portion B1 and 10% of the radius of the core 132.

Therefore, even if the third portion B2 is bent toward the core, 90% or more of the diameter of the core 132 of the electrode assembly 130 is open to the outside. If the core 132 is not blocked, there is no difficulty in the electrolyte

injection process, and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the core 132, the welding process may be easily performed between the current collector of the negative electrode (or, positive electrode) and the battery housing (or, terminal).

The second uncoated portion 43b has the same structure as the first uncoated portion 43a. In one modification, the second uncoated portion 43b may have a conventional electrode structure or the electrode structure of other embodiments (modifications).

The ends 131 of the first uncoated portion 43a and the second uncoated portion 43b may be bent from the outer circumference of the electrode assembly 130 toward the core. At this time, the first portion B1 is substantially not bent.

When the third portion B2 and the second portion B3 include a plurality of segments, the bending stress is alleviated to prevent the uncoated portions 43a, 43b near the bending point from being torn or abnormally deformed. In addition, when the width and/or height and/or separation pitch of the segments is adjusted according to the numerical range of the above embodiment, the segments are bent toward the core and overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region.

Meanwhile, in the above embodiments (modifications), the ends of the first uncoated portion 43a and the second uncoated portion 43b may be bent from the core toward the outer circumference. In this case, it is preferable that the second portion B3 is designed as the segment skip region (1) in FIG. 17a) with no segment and not bent toward the outer circumference. In addition, the radial width of the second portion B3 may be equal to or greater than the bending length of the outermost uncoated portion (or, segment) of the third portion B2. In this case, when the outermost uncoated portion (or, segment) of the third portion B2 is bent toward the outer circumference, the end of the bent portion does not protrude toward the inner surface of the battery housing beyond the outer circumference of the electrode assembly. In addition, the structural change pattern of the segments may be opposite to the above embodiments (modifications). For example, the heights of the segments may increase stepwise or gradually from the outer circumference toward the core. That is, by sequentially arranging the segment skip region (1) in FIG. 17a), the height variable region (2) in FIG. 17a), and the height uniform region (3) in FIG. 17a) from the outer circumference of the electrode assembly toward the core, in the bending surface region, the stack number uniform region in which the stack number of segments is 10 or more may appear in a desirable numerical range.

Various electrode assembly structures according to an embodiment of the present disclosure may be applied to a cylindrical battery.

Preferably, the cylindrical battery may be, for example, a cylindrical battery whose form factor ratio (defined as a value obtained by dividing the diameter of the cylindrical battery by height, namely a ratio of diameter (Φ) to height (H)) is greater than about 0.4. Here, the form factor means a value indicating the diameter and height of a cylindrical battery.

Preferably, the cylindrical battery may have a diameter of 40 mm to 50 mm and a height of 60 mm to 130 mm. The form factor of the cylindrical battery according to an embodiment of the present disclosure may be, for example, 46110, 4875, 48110, 4880, 4680, 4695. In the numerical value representing the form factor, first two numbers indi-

cate the diameter of the battery, and the remaining numbers indicate the height of the battery. The winding turn of the electrode assembly may be 50 turns to 60 turns.

When an electrode assembly having a tab-less structure is applied to a cylindrical battery having a form factor ratio of more than 0.4, the stress applied in the radial direction when the uncoated portion is bent is large, so that the uncoated portion may be easily torn. In addition, when welding the current collector to the bending surface region of the uncoated portion, it is necessary to sufficiently increase the number of stacked layers of the uncoated portion in the bending surface region in order to sufficiently secure the welding strength and lower the resistance. This requirement may be achieved by the electrode and the electrode assembly according to the embodiments (modifications) of the present disclosure.

A battery according to an embodiment of the present disclosure may be an approximately cylindrical battery, whose diameter is approximately 46 mm, height is approximately 110 mm, and form factor ratio is 0.418.

A battery according to another embodiment may be an approximately cylindrical battery, whose diameter is about 48 mm, height is about 75 mm, and form factor ratio is 0.640.

A battery according to still another embodiment may be an approximately cylindrical battery, whose diameter is approximately 48 mm, height is approximately 110 mm, and form factor ratio is 0.436.

A battery according to still another embodiment may be an approximately cylindrical battery, whose diameter is approximately 48 mm, height is approximately 80 mm, and form factor ratio is 0.600.

A battery according to still another embodiment may be an approximately cylindrical battery, whose diameter is approximately 46 mm, height is approximately 80 mm, and form factor ratio is 0.575.

A battery according to still another embodiment may be an approximately cylindrical battery, whose diameter is approximately 46 mm, height is approximately 95 mm, and form factor ratio is 0.484.

Conventionally, batteries having a form factor ratio of about 0.4 or less have been used. That is, conventionally, for example, 1865 battery, 2170 battery, etc. were used. The 1865 battery has a diameter of approximately 18 mm, height of approximately 65 mm, and a form factor ratio of 0.277. The 2170 battery has a diameter of approximately 21 mm, a height of approximately 70 mm, and a form factor ratio of 0.300.

Hereinafter, the cylindrical battery according to an embodiment of the present disclosure will be described in detail.

FIG. 24 is a cross-sectional view showing a cylindrical battery 140 according to an embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 24, the cylindrical battery 140 according to an embodiment of the present disclosure includes an electrode assembly 141 having a first electrode, a separator and a second electrode, a battery housing 142 for accommodating the electrode assembly 141, and a sealing body 143 for sealing an open end of the battery housing 142.

The battery housing 142 is a cylindrical container with an opening at the top. The battery housing 142 is made of a conductive metal material such as aluminum, steel or stainless steel. A nickel coating layer may be formed on the surface of the battery housing 142. The battery housing 142

accommodates the electrode assembly 141 in the inner space through the top opening and also accommodates the electrolyte.

The electrolyte may be a salt having a structure like A^+B^- . Here, A^+ includes an alkali metal cation such as Lit, Na^+ , or K^+ , or a combination thereof, and B^- includes at least one anion selected from the group consisting of F^- , Cl^- , Br^- , I^- , NO_3^- , $N(CN)_2^-$, BF_4^- , ClO_4^- , AlO_4^- , $AlCl_4^-$, PF_6^- , SbF_6^- , AsF_6^- , $BF_2C_2O_4^-$, $BC_4O_8^-$, $(CF_3)_2PF_4^-$, $(CF_3)_3PF_3^-$, $(CF_3)_4PF_2^-$, $(CF_3)_5PF^-$, $(CF_3)_6P^-$, $CF_3SO_3^-$, $C_4F_9SO_3^-$, $CF_3CF_2SO_3^-$, $(CF_3SO_2)_2N^-$, $(FSO_2)_2N^-$, $CF_3CF_2(CF_3)_2CO^-$, $(CF_3SO_2)_2CH^-$, $(SF_5)_3C^-$, $(CF_3SO_2)_3C^-$, $CF_3(CF_2)_7SO_3^-$, $CF_3CO_2^-$, $CH_3CO_2^-$, SCN^- and $(CF_3CF_2SO_2)_2N^-$.

The electrolyte may also be dissolved in an organic solvent. The organic solvent may employ propylene carbonate (PC), ethylene carbonate (EC), diethyl carbonate (DEC), dimethyl carbonate (DMC), dipropyl carbonate (DPC), dimethyl sulfoxide, acetonitrile, dimethoxyethane, diethoxyethane, tetrahydrofuran, N-methyl-2-pyrrolidone (NMP), ethyl methyl carbonate (EMC), γ -butyrolactone, or a mixture thereof.

The electrode assembly 141 may have a jelly-roll shape, but the present disclosure is not limited thereto. The electrode assembly 141 may be manufactured by winding a laminate formed by sequentially laminating a lower separator, a first electrode, an upper separator, and a second electrode at least once, based on the winding axis C, as shown in FIG. 2.

The first electrode and the second electrode have different polarities. That is, if one has positive polarity, the other has negative polarity. At least one of the first electrode and the second electrode may have an electrode structure according to the above embodiments (modifications). In addition, the other of the first electrode and the second electrode may have a conventional electrode structure or an electrode structure according to embodiments (modifications). The electrode pair included in the electrode assembly 141 is not limited to one electrode pair, two or more electrode pairs may be included.

The first uncoated portion 146a of the first electrode and the second uncoated portion 146b of the second electrode protrude at the upper portion and the lower portion of the electrode assembly 141, respectively. The first electrode has the electrode structure of the first embodiment (modification). Therefore, in the first uncoated portion 146a, the height of the second portion B3 is smaller than the height of the uncoated portion of the other region. The second portion B3 is spaced apart from the inner circumference of the battery housing 142, particularly the beading portion 147, by a predetermined distance. Therefore, the second portion B3 of the first electrode does not come into contact with the battery housing 142 electrically connected to the second electrode, thereby preventing an internal short circuit of the battery 140.

The second uncoated portion 146b of the second electrode may have the same structure as the first uncoated portion 146a. In another modification, the second uncoated portion 146b may selectively have the uncoated portion structures of the electrodes according to the embodiments (modifications).

The sealing body 143 may include a cap 143a having a plate shape, a first gasket 143b for providing airtightness between the cap 143a and the battery housing 142 and having insulation, and a connection plate 143c electrically and mechanically coupled to the cap 143a.

The cap 143a is a component made of a conductive metal material, and covers the top opening of the battery housing

142. The cap 143a is electrically connected to the uncoated portion 146a of the first electrode, and is electrically insulated from the battery housing 142 by means of the first gasket 143b. Accordingly, the cap 143a may function as the first electrode (for example, positive electrode) of the cylindrical battery 140.

The cap 143a is placed on the beading portion 147 formed on the battery housing 142, and is fixed by a crimping portion 148. Between the cap 143a and the crimping portion 148, the first gasket 143b may be interposed to secure the airtightness of the battery housing 142 and the electrical insulation between the battery housing 142 and the cap 143a. The cap 143a may have a protrusion 143d protruding upward from the center thereof.

The battery housing 142 is electrically connected to the second uncoated portion 146b of the second electrode. Therefore, the battery housing 142 has the same polarity as the second electrode. If the second electrode has negative polarity, the battery housing 142 also has negative polarity.

The battery housing 142 includes the beading portion 147 and the crimping portion 148 at the top thereof. The beading portion 147 is formed by press-fitting the periphery of the outer circumferential surface of the battery housing 142. The beading portion 147 prevents the electrode assembly 141 accommodated inside the battery housing 142 from escaping through the top opening of the battery housing 142, and may function as a support portion on which the sealing body 143 is placed.

The inner circumference of the beading portion 147 may be spaced apart from the second portion B3 of the first electrode by a predetermined interval. More specifically, the lower end of the inner circumference of the beading portion 147 is separated from the second portion B3 of the first electrode by a predetermined interval. In addition, since the second portion B3 has a low height, the second portion B3 is not substantially affected even when the battery housing 142 is press-fitted at the outside to form the beading portion 147. Therefore, the second portion B3 is not pressed by other components such as the beading portion 147, and thus partial shape deformation of the electrode assembly 141 is prevented, thereby preventing a short circuit inside the cylindrical battery 140.

Preferably, when the press-fit depth of the beading portion 147 is defined as D1 and the radial length from the inner circumference of the battery housing 142 to the boundary point between the second portion B3 and the third portion B2 is defined as D2, the relational expression $D1 \leq D2$ may be satisfied. In this case, when press-fitting the battery housing 142 to form the beading portion 147, it is possible to substantially prevent the second portion B3 from being damaged.

The crimping portion 148 is formed on the beading portion 147. The crimping portion 148 has an extended and bent shape to cover the outer circumference of the cap 143a disposed on the beading portion 147 and a part of the upper surface of the cap 143a.

The cylindrical battery 140 may further include a first current collector 144 and/or a second current collector 145 and/or an insulator 146.

The first current collector 144 is coupled to the upper portion of the electrode assembly 141. The first current collector 144 is made of a conductive metal material such as aluminum, copper, steel, nickel and so on, and is electrically connected to the first uncoated portion 146a of the first electrode. The electric connection may be made by welding. A lead 149 may be connected to the first current collector 144. The lead 149 may extend upward above the electrode

assembly 141 and be coupled to the connection plate 143c or directly coupled to the lower surface of the cap 143a. The lead 149 may be connected to other components by welding.

Preferably, the first current collector 144 may be integrally formed with the lead 149. In this case, the lead 149 may have an elongated plate shape extending outward from near the center of the first current collector 144.

The first current collector 144 may have a plurality of irregularities (not shown) formed radially on the lower surface thereof. When the radial irregularities are provided, the irregularities may be press-fitted into the first uncoated portion 146a of the first electrode by pressing the first current collector 144.

The first current collector 144 is coupled to an end of the first uncoated portion 146a. The first uncoated portion 146a and the first current collector 144 may be coupled by, for example, laser welding. Laser welding may be performed by partially melting a base material of the current collector 144. In one modification, the first current collector 144 and the first uncoated portion 146a may be welded with a solder interposed therebetween. In this case, the solder may have a lower melting point compared to the first current collector 144 and the first uncoated portion 146a. The laser welding may be replaced with resistance welding, ultrasonic welding, spot welding, or the like.

The second current collector 145 may be coupled to the lower surface of the electrode assembly 141. One side of the second current collector 145 may be coupled by welding to the second uncoated portion 146b, and the other side may be coupled to the inner bottom surface of the battery housing 142 by welding. The coupling structure between the second current collector 145 and the second uncoated portion 146b may be substantially the same as the coupling structure between the first current collector 144 and the first uncoated portion 146a.

The welding of the first current collector 144 and/or the second current collector 145 may prevent the electrodes from rotating when the electrode assembly 141 swells. In other words, the welding of the first current collector 144 and/or the second current collector 145 may enhance the effect of the winding structure of the electrode assembly to prevent or alleviate the collapse of the core. This effect appears identically in embodiments described later.

The uncoated portions 146a, 146b are not limited to the illustrated structure. Accordingly, the uncoated portions 146a, 146b may selectively have a conventional uncoated portion structure as well as the electrode uncoated portion structure according to the embodiments (modifications).

The insulator 146 may cover the first current collector 144. The insulator 146 may cover the first current collector 144 at the upper surface of the first current collector 144, thereby preventing direct contact between the first current collector 144 and the inner circumference of the battery housing 142.

The insulator 146 has a lead hole 151 so that the lead 149 extending upward from the first current collector 144 may be withdrawn therethrough. The lead 149 is drawn upward through the lead hole 151 and coupled to the lower surface of the connection plate 143c or the lower surface of the cap 143a.

A peripheral region of the edge of the insulator 146 may be interposed between the first current collector 144 and the beading portion 147 to fix the coupled body of the electrode assembly 141 and the first current collector 144. Accordingly, the movement of the coupled body of the electrode assembly 141 and the first current collector 144 may be

restricted in the winding axis direction Y of the battery 140, thereby improving the assembly stability of the battery 140.

The insulator 146 may be made of an insulating polymer resin. In one example, the insulator 146 may be made of polyethylene, polypropylene, polyimide, or polybutylene terephthalate.

The battery housing 142 may further include a venting portion 152 formed at a lower surface thereof. The venting portion 152 corresponds to a region having a smaller thickness compared to the peripheral region of the lower surface of the battery housing 142. The venting portion 152 is structurally weak compared to the surrounding area. Accordingly, when an abnormality occurs in the cylindrical battery 140 and the internal pressure increases to a predetermined level or more, the venting portion 152 may be ruptured so that the gas generated inside the battery housing 142 is discharged to the outside. The internal pressure at which the venting portion 152 is ruptured may be approximately 15 kgf/cm² to 35 kgf/cm².

The venting portion 152 may be formed continuously or discontinuously while drawing a circle at the lower surface of the battery housing 142. In one modification, the venting portion 152 may be formed in a straight pattern or other patterns.

FIG. 25 is a cross-sectional view showing a cylindrical battery 150 according to an embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 25, the cylindrical battery 150 has substantially the same configuration as the cylindrical battery 140 of FIG. 24, except that the electrode structure of the second embodiment (modification) is adopted in the first uncoated portion 146a of the first electrode.

Referring to FIG. 25, the first uncoated portion 146a of the first electrode may have a shape in which the height of the second portion B3 gradually or stepwise decreases toward the inner circumference of the battery housing 142. Preferably, an imaginary line connecting the top of the second portion B3 may have the same or similar shape as the inner circumference of the beading portion 147.

The second portion B3 forms an inclined surface. Therefore, when the battery housing 142 is press-fitted to form the beading portion 147, the second portion B3 may be prevented from being compressed and damaged by the beading portion 147. In addition, it is possible to suppress a phenomenon in which the second portion B3 contacts the battery housing 142 having a different polarity and causes an internal short circuit.

The other configuration of the cylindrical battery 150 is substantially the same as the embodiment (modification) described above.

The uncoated portions 146a, 146b are not limited to the illustrated structure. Accordingly, the uncoated portions 146a, 146b may selectively have a conventional uncoated portion structure as well as the electrode uncoated portion structure according to the embodiments (modifications).

FIG. 26 is a cross-sectional view showing a cylindrical battery 160 according to still another embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 26, the cylindrical battery 160 is substantially the same as the cylindrical batteries 140, 150, described above, except that the lead 149 connected to the first current collector 144 is connected directly to the cap 143a of the sealing body 143 through the lead hole 151 of the insulator 146, and the insulator 146 and the first current collector 144 have a structure in close contact with the lower surface of the cap 143a.

In the cylindrical battery 160, the diameter of the first current collector 144 and the diameter of the outermost side of the third portion B2 are smaller than the minimum inner diameter of the battery housing 142. Also, the diameter of the first current collector 144 may be equal to or larger than the diameter of the outermost side of the third portion B2.

Specifically, the minimum inner diameter of the battery housing 142 may correspond to the inner diameter of the battery housing 142 at a position where the beading portion 147 is formed. At this time, the outermost diameters of the first current collector 144 and the third portion B2 are smaller than the inner diameter of the battery housing 142 at the position where the beading portion 147 is formed. Also, the diameter of the first current collector 144 may be equal to or larger than the outermost diameter of the third portion B2. The peripheral region of the edge of the insulator 146 may be interposed between the second portion B3 and the beading portion 147 in a state of being bent downward to fix the coupled body of the electrode assembly 141 and the first current collector 144.

Preferably, the insulator 146 includes a portion covering the second portion B3 and a portion covering the first current collector 144, and a portion connecting the two portions may have a curved shape corresponding to the curved shape of the beading portion 147. The insulator 146 may insulate the inner circumference of the second portion B3 and the beading portion 147 and at the same time insulate the first current collector 144 and the inner circumference of the beading portion 147.

The first current collector 144 may be located higher than the lower end of the beading portion 147 and may be coupled to the first portion B1 and the third portion B2. At this time, the press-fit depth D1 of the beading portion 147 is less than or equal to the distance D2 from the inner circumference of the battery housing 142 to the boundary between the second portion B3 and the third portion B2. Accordingly, the first portion B1 and the third portion B2 as well as the first current collector 144 coupled thereto may be located higher than the lower end of the beading portion 147. The lower end of the beading portion 147 means a bending point B between the portion of the battery housing 142 where the electrode assembly 141 is accommodated and the beading portion 147.

Since the first portion B1 and the third portion B2 occupy the inner space of the beading portion 147 in the radial direction, the empty space between the electrode assembly 141 and the cap 143a may be minimized. In addition, the connection plate 143c located in the empty space between the electrode assembly 141 and the cap 143a is omitted. Therefore, the lead 149 of the first current collector 144 may be directly coupled to the lower surface of the cap 143a. According to the above structure, the empty space in the battery is reduced, and the energy density may be maximized as much as the reduced empty space.

In the cylindrical battery 160, the first current collector 144 and the second current collector 145 may be welded to the ends of the first and second uncoated portions 146a, 146b, respectively, in the same manner as in the above embodiment.

The uncoated portions 146a, 146b are not limited to the illustrated structure. Accordingly, the uncoated portions 146a, 146b may selectively have a conventional uncoated portion structure as well as the electrode uncoated portion structure according to the embodiments (modifications).

FIG. 27 is a cross-sectional view showing a cylindrical battery 170 according to an embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 27, the structure of the electrode assembly of the cylindrical battery 170 is substantially the same as that of the cylindrical battery 140 of in FIG. 17, and the other structure except for the electrode assembly is changed.

Specifically, the cylindrical battery 170 includes a battery housing 171 through which a terminal 172 is installed. The terminal 172 is installed through a perforation hole formed in the closed surface (the upper surface in the drawing) of the battery housing 171. The terminal 172 is riveted to the perforation hole of the battery housing 171 in a state where a second gasket 173 made of an insulating material is interposed therebetween. The terminal 172 is exposed to the outside in a direction opposite to the direction of gravity.

The terminal 172 includes a terminal exposing portion 172a and a terminal insert portion 172b. The terminal exposing portion 172a is exposed to the outside of the closed surface of the battery housing 171. The terminal exposing portion 172a may be located approximately at a central portion of the closed surface of the battery housing 171. The maximum diameter of the terminal exposing portion 172a may be larger than the maximum diameter of the perforation hole formed in the battery housing 171. The terminal insert portion 172b may be electrically connected to the uncoated portion 146a of the first electrode through approximately the central portion of the closed surface of the battery housing 171. The lower edge of the terminal insert portion 172b may be riveted onto the inner surface of the battery housing 171. That is, the lower edge of the terminal insert portion 172b may have a shape curved toward the inner surface of the battery housing 171. A flat portion 172c is included at the inner side of the lower edge of the terminal insert portion 172b. The maximum diameter of the lower portion of the riveted terminal insert portion 172b may be larger than the maximum diameter of the perforation hole of the battery housing 171.

The flat portion 172c of the terminal insert portion 172b may be welded to the center portion of the first current collector 144 connected to the first uncoated portion 146a of the first electrode. The welding may be replaced with other welding methods such as ultrasonic welding.

An insulator 174 made of an insulating material may be interposed between the first current collector 144 and the inner surface of the battery housing 171. The insulator 174 covers the upper portion of the first current collector 144 and the top edge of the electrode assembly 141. Accordingly, it is possible to prevent the second portion B3 of the electrode assembly 141 from contacting the inner surface of the battery housing 171 having a different polarity to cause a short circuit.

The thickness of the insulator 174 corresponds to or is slightly greater than the distance between the upper surface of the first current collector 144 and the inner surface of the closed portion of the battery housing 171. Accordingly, the insulator 174 may contact the upper surface of the first current collector 144 and the inner surface of the closed portion of the battery housing 171.

The terminal insert portion 172b of the terminal 172 may be welded to the first current collector 144 through the perforation hole of the insulator 174. A diameter of the perforation hole formed in the insulator 174 may be larger than a diameter of the riveting portion at the lower end of the terminal insert portion 172b. Preferably, the perforation hole may expose the lower portion of the terminal insert portion 172b and the second gasket 173.

The second gasket 173 is interposed between the battery housing 171 and the terminal 172 to prevent the battery housing 171 and the terminal 172 having opposite polarities

from electrically contacting each other. Accordingly, the upper surface of the battery housing 171 having an approximately flat shape may function as the second electrode (for example, negative electrode) of the cylindrical battery 170.

The second gasket 173 includes a gasket exposing portion 173a and a gasket insert portion 173b. The gasket exposing portion 173a is interposed between the terminal exposing portion 172a of the terminal 172 and the battery housing 171. The gasket insert portion 173b is interposed between the terminal insert portion 172b of the terminal 172 and the battery housing 171. The gasket insert portion 173b may be deformed together when the terminal insert portion 172b is riveted, so as to be in close contact with the inner surface of the battery housing 171. The second gasket 173 may be made of, for example, a polymer resin having insulation property.

The gasket exposing portion 173a of the second gasket 173 may have an extended shape to cover the outer circumference of the terminal exposing portion 172a of the terminal 172. When the second gasket 173 covers the outer circumference of the terminal 172, it is possible to prevent a short circuit from occurring while an electrical connection part such as a bus bar is coupled to the upper surface of the battery housing 171 and/or the terminal 172. Although not shown in the drawings, the gasket exposing portion 173a may have an extended shape to cover not only the outer circumference surface of the terminal exposing portion 172a but also a part of the upper surface thereof.

When the second gasket 173 is made of a polymer resin, the second gasket 173 may be coupled to the battery housing 171 and the terminal 172 by thermal fusion. In this case, airtightness at the coupling interface between the second gasket 173 and the terminal 172 and at the coupling interface between the second gasket 173 and the battery housing 171 may be enhanced. Meanwhile, when the gasket exposing portion 173a of the second gasket 173 has a shape extending to the upper surface of the terminal exposing portion 172a, the terminal 172 may be integrally coupled with the second gasket 173 by insert injection molding.

In the upper surface of the battery housing 171, a remaining area or part 175 other than the area occupied by the terminal 172 and the second gasket 173 corresponds to the second electrode terminal having a polarity opposite to that of the terminal 172.

The second current collector 176 is coupled to the lower portion of the electrode assembly 141. The second current collector 176 is made of a conductive metal material such as aluminum, steel, copper or nickel, and is electrically connected to the second uncoated portion 146b of the second electrode.

Preferably, the second current collector 176 is electrically connected to the battery housing 171. To this end, at least a portion of the edge of the second current collector 176 may be interposed and fixed between the inner surface of the battery housing 171 and a first gasket 178b. In one example, at least a portion of the edge of the second current collector 176 may be fixed to the beading portion 180 by welding in a state of being supported on the lower surface of the beading portion 180 formed at the bottom of the battery housing 171. In one modification, at least a portion of the edge of the second current collector 176 may be directly welded to the inner wall surface of the battery housing 171.

The second current collector 176 may include a plurality of irregularities (not shown) radially formed on a surface facing the second uncoated portion 146b. When the irregu-

larities are formed, the irregularities may be press-fitted into the second uncoated portion 146b by pressing the second current collector 176.

Preferably, the second current collector 176 and the second uncoated portion 146b may be coupled by welding, for example laser welding. In addition, the welded portion of the second current collector 176 and the second uncoated portion 146b may be spaced apart by a predetermined interval toward the core C based on the inner circumference of the beading portion 180.

A sealing body 178 for sealing the lower open end of the battery housing 171 includes a cap 178a having a plate shape and a first gasket 178b. The first gasket 178b electrically separates the cap 178a and the battery housing 171. A crimping portion 181 fixes the edge of the cap 178a and the first gasket 178b together. The cap 178a has a venting portion 179. The configuration of the venting portion 179 is substantially the same as the above embodiment (modification). The lower surface of the cap 178a may be located above the lower end of the crimping portion 181. In this case, a space is formed under the cap 178a to smoothly perform venting. In particular, it is useful when the cylindrical battery 170 is installed so that the crimping portion 181 faces the direction of gravity.

Preferably, the cap 178a is made of a conductive metal material. However, since the first gasket 178b is interposed between the cap 178a and the battery housing 171, the cap 178a does not have electrical polarity. The sealing body 178 seals the open end of the lower portion of the battery housing 171 and mainly functions to discharge gas when the internal pressure of the battery 170 increases over a critical value. A threshold value of the pressure is 15 kgf/cm² to 35 kgf/cm².

Preferably, the terminal 172 electrically connected to the first uncoated portion 146a of the first electrode is used as the first electrode terminal. In addition, in the upper surface of the battery housing 171 electrically connected to the second uncoated portion 146b of the second electrode through the second current collector 176, a part 175 except for the terminal 172 is used as the second electrode terminal having a different polarity from the first electrode terminal. If two electrode terminals are located at the upper portion of the cylindrical battery 170 as above, it is possible to arrange electrical connection components such as bus bars at only one side of the cylindrical battery 170. This may bring about simplification of the battery pack structure and improvement of energy density. In addition, since the part 175 used as the second electrode terminal has an approximately flat shape, a sufficient connection area may be secured for connecting electrical connection components such as bus bars. Accordingly, the cylindrical battery 170 may reduce the resistance at the connection portion of the electrical connection components to a desirable level.

Meanwhile, the structure of the electrode assembly 141 and the structure of the uncoated portion are not limited to those shown, and may be replaced with structures of the above embodiments (modifications).

FIG. 28 is a cross-sectional view showing a cylindrical battery 180 according to still another embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 28, the electrode assembly 141 of the cylindrical battery 180 has substantially the same structure as that of the cylindrical battery 150 shown in FIG. 25, and the other configuration than the electrode assembly 141 is substantially the same as that of the cylindrical battery 170 shown in FIG. 27.

Accordingly, the configuration of the embodiments (modifications) of the cylindrical batteries 150, 170 may be equally applied to the cylindrical battery 180.

In addition, the structure of the electrode assembly 141 and the structure of the uncoated portion are not limited to those shown, and may be replaced with structures of the above embodiments (modifications).

FIG. 29 is a cross-sectional view showing a cylindrical battery 190 according to still another embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 29, the cylindrical battery 190 includes the electrode assembly 110 shown in FIG. 21, and other configurations except for the electrode assembly 110 are substantially the same as those of the cylindrical battery 140 shown in FIG. 24. Accordingly, the configuration described with reference to FIGS. 21 and 24 may be substantially equally applied to this embodiment.

Referring to FIGS. 17a and 29, the first and second uncoated portions 146a, 146b of the electrode assembly 110 are bent in the radial direction of the electrode assembly 110, for example from the outer circumference toward the core, to form a bending surface region F.

The first portion B1 has a lower height than the other portion and corresponds to segment skip region a1 with no segment, so it is not bent toward the core.

Preferably, the bending surface region F may include the segment skip region a1, the segment height variable region a2, and the segment height uniform region a3 from the core toward the outer circumference.

As shown in FIGS. 17c, 17d, and 17e, the bending surface region F includes a stack number uniform region b1 having a stack number of 10 or more adjacent to the segment skip region a1.

The bending surface region F may also include a stack number decrease region b2 adjacent to the outer circumference of the electrode assembly 110, where the stack number of segments decreases toward the outer circumference. Preferably, the stack number uniform region b1 may be set as a welding target area.

In the bending surface region F, the preferred numerical ranges of the ratio (a2/c) of the height variable region a2 to the radial region c containing segments, the ratio (b1/c) of the stack number uniform region b1 to the radial region c containing segments, and the ratio of the area of the stack number uniform region b1 to the area of the bending surface region F are already described above, and thus will not be described again.

The first current collector 144 may be laser-welded to the bending surface region F of the first uncoated portion 146a, and the second current collector 145 may be laser-welded to the bending surface region F of the second uncoated portion 146b. The welding method may be replaced by ultrasonic welding, resistance welding, spot welding, and the like.

Preferably, an area of 50% or more of the welding regions W of the first current collector 144 and the second current collector 145 may overlap with the stack number uniform region b1 of the bending surface region F. Optionally, the remaining area of the welding region W may overlap with the stack number decrease region b2 of bending surface region F. In terms of high welding strength, low resistance of the welding interface, and prevention of damage to the separator or the active material layer, it is more preferable that the entire welding region W overlaps the stack number uniform region b1.

Preferably, in the stack number uniform region b1 and, optionally, the stack number decrease region b2 overlapping with the welding region W, the stack number of segments may be 10 to 35.

Optionally, when the segment stack number of the stack number decrease region b2 overlapping with the welding region W is less than 10, the laser output for welding of the stack number decrease region b2 may be lowered than the laser output for welding of the stack number uniform region b1. That is, when the welding region W overlaps with the stack number uniform region b1 and the stack number decrease region b2 at the same time, the laser output may be varied according to the stack number of segments. In this case, the welding strength of the stack number uniform region b1 may be greater than the welding strength of the stack number decrease region b2.

In the bending surface region F formed on the upper portion and the lower portion of the electrode assembly 110, the radial length of the segment skip region a1 and/or the segment height variable region a2 and/or the segment height uniform region a3 may be the same or different.

In the electrode assembly 110, the first portion B1 has a relatively smaller height than other portions. In addition, as shown in FIG. 21, the bending length H of the uncoated portion located at the innermost side is smaller than the sum of the radial length R of the first portion B1 and 10% of the radius of the core 112.

Therefore, even if the first uncoated portion 146a is bent toward the core, 90% or more of the diameter of the core 112 of the electrode assembly 110 may be open to the outside. If the core 112 is not blocked, there is no difficulty in the electrolyte injection process, and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the core 112, the welding process may be easily performed between the second current collector 145 and the battery housing 142.

In the case where the uncoated portions 146a, 146b have a segment structure, if the width and/or height and/or separation pitch of the segments is adjusted to satisfy the numerical range of the above embodiment, when the segments are bent, the segments are overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region F.

Preferably, the first current collector 144 and the second current collector 145 may have outer diameters covering the end of the segment 61, 61' (FIG. 17f) bent at the last winding turn of the height uniform region a3 of the first electrode and the second electrode. In this case, welding is possible in a state while the segments forming the bending surface region F are uniformly pressed by the current collector, and the tightly stacked state of the segments may be well maintained even after welding. The tightly stacked state means a state where there is substantially no gap between the segments as shown in FIG. 17a. The tightly stacked state contributes to lowering the resistance of the cylindrical battery 190 to a level suitable for rapid charging (for example, 4 milliohms) or less.

If the first current collector 144 and/or the second current collector 145 is welded to the surface region F, it is possible to prevent the electrodes from rotating when the electrode assembly 110 swells. In other words, the welding of the first current collector 144 and/or the second current collector 145 may enhance the effect of the winding structure of the electrode assembly to prevent or alleviate the collapse of the core.

The structure of the uncoated portions 146a, 146b may be changed to the structure according to the above embodi-

ments (modifications). In addition, a conventional uncoated portion structure may be applied to any one of the uncoated portions 146a, 146b without limitation.

FIG. 30 is a cross-sectional view showing a cylindrical battery 200 according to still another embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 30, the cylindrical battery 200 includes the electrode assembly 110 shown in FIG. 21, and other configurations except for the electrode assembly 110 are substantially the same as those of the cylindrical battery 180 shown in FIG. 28. Accordingly, the configuration described with reference to FIGS. 21 and 28 may be substantially equally applied to this embodiment.

As shown in FIGS. 17a and 30, the first and second uncoated portions 146a, 146b of the electrode assembly 110 are bent in the radial direction of the electrode assembly 110, for example from the outer circumference toward the core, to form the bending surface region F.

The first portion B1 has a lower height than the other portion and corresponds to the segment skip region a1 with no segment, so it is not bent toward the core.

Preferably, the bending surface region F may include a segment skip region a1, a segment height variable region a2, and a segment height uniform region a3 from the core toward the outer circumference.

As shown in FIGS. 17c, 17d and 17e, the bending surface region F includes a stack number uniform region b1 having the stack number of 10 or more adjacent to the segment skip region a1.

The bending surface region F may also include a stack number decrease region b2 adjacent to the outer circumference of the electrode assembly 110, where the stack number of segments decreases toward the outer circumference. Preferably, the stack number uniform region b1 may be set as a welding target area.

In the bending surface region F, the preferred numerical ranges of the ratio (a2/c) of the segment height variable region a2 to the radial region c including the segments, the ratio (b1/c) of the segment stack number uniform region b1 to the radial region c including the segments, and the ratio of the area of the stack number uniform region b1 to the area of the bending surface region F are already described above and thus will not be described again.

The first current collector 144 may be laser-welded to the bending surface region F of the first uncoated portion 146a, and the second current collector 176 may be laser-welded to the bending surface region F of the second uncoated portion 146b. The welding method may be replaced by ultrasonic welding, resistance welding, spot welding, and the like. The welding region W between the second current collector 176 and the second uncoated portion 146b may be spaced apart from the inner surface of the beading portion 180 by a predetermined interval.

Preferably, an area of 50% or more of the welding regions W of the first current collector 144 and the second current collector 176 may overlap with the stack number uniform region b1 of the bending surface region F. Optionally, the remaining area of the welding region W may overlap with the stack number decrease region b2 of the bending surface region F. In terms of high welding strength, low resistance of the welding interface, and prevention of damage to the separator or the active material layer, it is more preferable that the entire welding region W overlaps with the stack number uniform region b1.

Preferably, in the stack number uniform region b1 and, optionally, the stack number decrease region b2 overlapping with the welding region W, the stack number of segments may be 10 to 35.

Optionally, when the segment stack number of the stack number decrease region b2 overlapping with the welding region W is less than 10, the laser output for welding of the stack number decrease region b2 may be lowered than the laser output for welding of the stack number uniform region b1. That is, when the welding region W overlaps with the stack number uniform region b1 and the stack number decrease region b2 at the same time, the laser output may be varied according to the stack number of segments. In this case, the welding strength of the stack number uniform region b1 may be greater than the welding strength of the stack number decrease region b2.

In the bending surface region F formed on the upper portion and the lower portion of the electrode assembly 110, the radial length of the segment skip region a1 and/or the segment height variable region a2 and/or the segment height uniform region a3 may be the same or different.

In the electrode assembly 110, the first portion B1 has a relatively smaller height than other portions. In addition, as shown in FIG. 21, the bending length H of the segment located at the innermost side is smaller than the sum of the radial length R of the first portion B1 and 10% of the radius of the core 112.

Therefore, even if the uncoated portion 146a is bent toward the core, 90% or more of the diameter of the core 112 of the electrode assembly 110 may be open to the outside. If the core 112 is not blocked, there is no difficulty in the electrolyte injection process, and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the core 112, the welding process may be easily performed between the first current collector 144 and the terminal 172.

In the case where the uncoated portions 146a, 146b have a segment structure, if the width and/or height and/or separation pitch of the segments is adjusted to satisfy the numerical range of the above embodiment, when the segments are bent, the segments are overlapped in several layers enough to secure sufficient welding strength, and an empty hole (gap) is not formed in the bending surface region F.

Preferably, the area of the first current collector 144 and the second current collector 176 in contact with the first and second uncoated portions 146a, 146b may have outer diameters covering the end of the segment 61, 61' (FIG. 17f) bent at the last winding turn of the height uniform region a3 of the first electrode and the second electrode. In this case, welding is possible in a state while the segments forming the bending surface region F are uniformly pressed by the current collector, and the tightly stacked state of the segments may be well maintained even after welding. The tightly stacked state means a state where there is substantially no gap between the segments as shown in FIG. 17a. The tightly stacked state contributes to lowering the resistance of the cylindrical battery 200 to a level suitable for rapid charging (for example, 4 milliohms) or less.

If the first current collector 144 and/or the second current collector 145 is welded to the surface region F, it is possible to prevent the electrodes from rotating when the electrode assembly 110 swells. In other words, the welding of the first current collector 144 and/or the second current collector 145 may enhance the effect of the winding structure of the electrode assembly to prevent or alleviate the collapse of the core.

The structure of the uncoated portions 146a, 146b may be changed to the structure according to the above embodiments (modifications). In addition, a conventional uncoated portion structure may be applied to any one of the uncoated portions 146a, 146b without limitation.

FIG. 31 is a cross-sectional view showing a cylindrical battery 210 according to an embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 31, the cylindrical battery 210 includes the electrode assembly 100 shown in FIG. 20, and other components except for the electrode assembly 100 are substantially the same as those of the cylindrical battery 140 shown in FIG. 24. Accordingly, the configuration described with reference to FIGS. 20 and 24 may be substantially equally applied to this embodiment.

Preferably, the first and second uncoated portions 146a, 146b of the electrode assembly 100 are divided into a plurality of segments, and the segments are bent in the radial direction of the electrode assembly 100, for example from the outer circumference toward the core. At this time, since the first portion B1 of the first uncoated portion 146a and the second portion B3 have a lower height than the other portions and do not include segments, they are not substantially bent. This is also identical in the case of the second uncoated portion 146b.

Also in this embodiment, the bending surface region F may include a segment skip region a1, a segment height variable region a2, and a segment height uniform region a3 from the core toward the outer circumference. However, since the second portion B3 is not bent, the radial length of the bending surface region F may be shorter than in the case of the above embodiment.

As shown in FIGS. 17c, 17d and 17e, the bending surface region F includes a stack number uniform region b1 having the stack number of 10 or more adjacent to the segment skip region a1.

The bending surface region F may also include a stack number decrease region b2 adjacent to the second portion B3 of the electrode assembly 100, in which the stack number of segments decreases toward the outer circumference. Preferably, the stack number uniform region b1 may be set as a welding target area.

In the bending surface region F, the preferred numerical range of the ratio (a2/c) of the segment height variable region a2 to the radial region c including the segments, the ratio (b1/c) of the segment stack number uniform region b1 to the radial region c including the segments, and the ratio of the area of the stack number uniform region b1 to the area of the bending surface region F are already described above and thus will not be described again.

The first current collector 144 may be welded to the bending surface region F of the first uncoated portion 146a, and the second current collector 145 may be welded to the bending surface region F of the second uncoated portion 146b.

The overlapping relationship between the stack number uniform region b1 and the stack number decrease region b2 and the welding region W, the outer diameters of the first current collector 144 and the first current collector 145, and the configuration in which the first portion B1 does not block 10% or more of the diameter of the core are substantially the same as described above.

Meanwhile, the second portion B3 does not include segments, and the height of the second portion B3 is lower than that of the third portion B2. Therefore, when the first uncoated portion 146a is bent, the second portion B3 is not substantially bent. In addition, since the second portion B3

is sufficiently spaced from the beading portion 147, the problem of damage to the second portion B3 may be solved while the beading portion 147 is press-fitted.

If the first current collector 144 and/or the second current collector 145 is welded to the surface region F, it is possible to prevent the electrodes from rotating when the electrode assembly 100 swells. In other words, the welding of the first current collector 144 and/or the second current collector 145 may enhance the effect of the winding structure of the electrode assembly to prevent or alleviate the collapse of the core.

The structure of the uncoated portions 146a, 146b may be changed to the structure according to the above embodiments (modifications). In addition, a conventional uncoated portion structure may be applied to any one of the uncoated portions 146a, 146b without limitation.

FIG. 32 is a cross-sectional view showing a cylindrical battery 220 according to an embodiment of the present disclosure, taken along the Y-axis direction.

Referring to FIG. 32, the cylindrical battery 220 includes the electrode assembly 100 shown in FIG. 31, and other components except for the electrode assembly 100 are substantially the same as those of the cylindrical battery 180 shown in FIG. 28. Accordingly, the configuration described with reference to FIGS. 28 and 31 may be substantially equally applied to this embodiment.

Preferably, the first and second uncoated portions 146a, 146b of the electrode assembly 100 are divided into a plurality of segments, and are bent from the outer circumference toward the core. At this time, since the first portion B1 of the first uncoated portion 146a and the second portion B3 have a lower height than the other portions and do not include segments, they are not substantially bent. This is also identical in the case of the second uncoated portion 146b.

Therefore, in this embodiment, similar to the embodiment of FIG. 31, the bending surface region F may include a segment skip region a1, a segment height variable region a2, and a segment height uniform region a3 from the core toward the outer circumference. However, since the second portion B3 is not bent, the radial length of the bending surface region F may be shorter than in the case of the above embodiment.

As shown in FIGS. 17c, 17d and 17e, the bending surface region F includes a stack number uniform region b1 having the stack number of 10 or more adjacent to the segment skip region a1.

The bending surface region F may also include a stack number decrease region b2 adjacent to the second portion B3 of the electrode assembly 110, in which the stack number of segments decreases toward the outer circumference. Preferably, the stack number uniform region b1 may be set as a welding target area.

In the bending surface region F, the preferred numerical range of the ratio (a2/c) of the segment height variable region a2 to the radial region c including the segments, the ratio (b1/c) of the segment stack number uniform region b1 to the radial region c including the segments, and the ratio of the area of the stack number uniform region b1 to the area of the bending surface region F are already described above and thus will not be described again.

The first current collector 144 may be welded to the bending surface region F of the first uncoated portion 146a, and the second current collector 176 may be welded to the bending surface region F of the second uncoated portion 146b.

The overlapping relationship between the stack number uniform region b1 and the stack number decrease region b2

and the welding region W, the outer diameters of the first current collector 144 and the first current collector 176, and the configuration in which the first portion B1 does not block 10% or more of the diameter of the core are substantially the same as described above.

If the first current collector 144 and/or the second current collector 145 is welded to the surface region F, it is possible to prevent the electrodes from rotating when the electrode assembly 100 swells. In other words, the welding of the first current collector 144 and/or the second current collector 145 may enhance the effect of the winding structure of the electrode assembly to prevent or alleviate the collapse of the core.

The structure of the uncoated portions 146a, 146b may be changed to the structure according to the above embodiments (modifications). In addition, a conventional uncoated portion structure may be applied to any one of the uncoated portions 146a, 146b without limitation.

In the embodiments (modifications), the first current collector 144 and the second current collector 176 included in the cylindrical batteries 170, 180, 200, 220 including the terminal 172 may have an improved structure as shown in FIGS. 33 and 34.

The improved structure of the first current collector 144 and the second current collector 176 may contribute to lowering the resistance of the cylindrical battery, improving vibration resistance, and improving energy density. In particular, the first current collector 144 and the second current collector 176 are more effective when used in a large cylindrical battery whose ratio of diameter to height is greater than 0.4.

FIG. 33 is a top plan view showing the structure of the first current collector 144 according to an embodiment of the present disclosure.

Referring to FIGS. 30 and 33 together, the first current collector 144 may include an edge portion 144a, a first uncoated portion coupling portion 144b, and a terminal coupling portion 144c. The edge portion 144a is disposed on the electrode assembly 110. The edge portion 144a may have a substantially rim shape having an empty space (S_{open}) formed therein. In the drawings of the present disclosure, only a case in which the edge portion 144a has a substantially circular rim shape is illustrated, but the present disclosure is not limited thereto. The edge portion 144a may have a substantially rectangular rim shape, a hexagonal rim shape, an octagonal rim shape, or other rim shapes, unlike the illustrated one. The number of the edge portion 144a may be increased to two or more. In this case, another edge portion in the form of a rim may be included inside the edge portion 144a.

The terminal coupling portion 144c may have a diameter equal to or greater than the diameter of the flat portion 172c formed on the bottom surface of the terminal 172 in order to secure a welding region for coupling with the flat portion 172c formed on the bottom surface of the terminal 172.

The first uncoated portion coupling portion 144b extends inward from the edge portion 144a and is coupled to the uncoated portion 146a by welding. The terminal coupling portion 144c is spaced apart from the first uncoated portion coupling portion 144b and is positioned inside the edge portion 144a. The terminal coupling portion 144c may be coupled to the terminal 172 by welding. The terminal coupling portion 144c may be located, for example, approximately at the center of the inner space (S_{open}) surrounded by the edge portion 144a. The terminal coupling portion 144c may be provided at a position corresponding to the hole formed in the core C of the electrode assembly 110. The

terminal coupling portion 144c may be configured to cover the hole formed in the core C of the electrode assembly 110 so that the hole formed in the core C of the electrode assembly 110 is not exposed out of the terminal coupling portion 144c. To this end, the terminal coupling portion 144c may have a larger diameter or width than the hole formed in the core C of the electrode assembly 110.

The first uncoated portion coupling portion 144b and the terminal coupling portion 144c may not be directly connected, but may be disposed to be spaced apart from each other and indirectly connected by the edge portion 144a. Since the first current collector 144 has a structure in which the first uncoated portion coupling portion 144b and the terminal coupling portion 144c are not directly connected to each other but are connected through the edge portion 144c as above, when shock and/or vibration occurs at the cylindrical battery 200, it is possible to disperse the shock applied to the coupling portion between the first uncoated portion coupling portion 144b and the first uncoated portion 146a and the coupling portion between the terminal coupling portion 144c and the terminal 172. In the drawings of the present disclosure, only a case in which four first uncoated portion coupling portions 144b are provided is illustrated, but the present disclosure is not limited thereto. The number of the first uncoated portion coupling portions 144b may be variously determined in consideration of manufacturing difficulty according to the complexity of the shape, electric resistance, the inner space (S_{open}) inside the edge portion 144a considering electrolyte impregnation, and the like.

The first current collector 144 may further include a bridge portion 144d extending inward from the edge portion 144a and connected to the terminal coupling portion 144c. At least a part of the bridge portion 144d may have a smaller sectional area compared to the first uncoated portion coupling portion 144b and the edge portion 144a. For example, at least a part of the bridge portion 144d may be formed to have a smaller width and/or thickness compared to the first uncoated portion coupling portion 144b. In this case, the electric resistance increases in the bridge portion 144d. Therefore, when a current flows through the bridge portion 144d, the relatively large resistance causes a part of the bridge portion 144d to be melted due to overcurrent heating. Accordingly, the overcurrent is irreversibly blocked. The sectional area of the bridge portion 144d may be adjusted to an appropriate level in consideration of the overcurrent blocking function.

The bridge portion 144d may include a taper portion 144e whose width is gradually decreased from the inner surface of the edge portion 144a toward the terminal coupling portion 144c. When the taper portion 144e is provided, the rigidity of the component may be improved at the connection portion between the bridge portion 144d and the edge portion 144a. When the taper portion 144e is provided, in the process of manufacturing the cylindrical battery 200, for example, a transfer device and/or a worker may easily and safely transport the first current collector 144 and/or a coupled body of the first current collector 144 and the electrode assembly 110 by gripping the taper portion 144e. That is, when the taper portion 144e is provided, it is possible to prevent product defects that may occur by gripping a portion where welding is performed with other components such as the first uncoated portion coupling portion 144b and the terminal coupling portion 144c.

The first uncoated portion coupling portion 144b may be provided in plural. The plurality of first uncoated portion coupling portions 144b may be disposed substantially at regular intervals from each other in the extending direction

of the edge portion 144a. An extension length of each of the plurality of first uncoated portion coupling portions 144b may be substantially equal to each other. The first uncoated portion coupling portion 144b may be coupled to the bending surface region F of the uncoated portion 146a by laser welding. The welding may be replaced by ultrasonic welding, spot welding, or the like.

A welding pattern 144f formed by welding between the first uncoated portion coupling portion 144b and the bending surface region F may have a structure extending along the radial direction of the electrode assembly 110. The welding pattern 144f may be an array of line patterns or dot patterns.

The welding pattern 144f corresponds to the welding region. Therefore, it is desirable that the welding pattern 144f overlaps with the stack number uniform region b1 of the bending surface region F by 50% or more. The welding pattern 144f that does not overlap with the stack number uniform region b1 may overlap with the stack number decrease region b2. More preferably, the entire welding pattern 144f may overlap with the stack number uniform region b1 of the bending surface region F. In the bending surface region F below the point where the welding pattern 144f is formed, the stack number uniform region b1 and, optionally, the stack number decrease region b2 preferably have the stack number of 10 or more.

The terminal coupling portion 144c may be disposed to be surrounded by the plurality of first uncoated portion coupling portions 144b. The terminal coupling portion 144c may be coupled to the flat portion 172c of the terminal 172 by welding. The bridge portion 144d may be positioned between a pair of first uncoated portion coupling portions 144b adjacent to each other. In this case, the distance from the bridge portion 144d to any one of the pair of first uncoated portion coupling portions 144b along the extending direction of the edge portion 144a may be substantially equal to the distance from the bridge portion 144d to the other one of the pair of first uncoated portion coupling portions 144b along the extending direction of the edge portion 144a. The plurality of first uncoated portion coupling portions 144b may be formed to have substantially the same sectional area. The plurality of first uncoated portion coupling portions 144b may be formed to have substantially the same width and thickness.

Although not shown in the drawings, the bridge portion 144d may be provided in plural. Each of the plurality of bridge portions 144d may be disposed between a pair of first uncoated portion coupling portions 144b adjacent to each other. The plurality of bridge portions 144d may be disposed substantially at regular intervals to each other in the extending direction of the edge portion 144a. A distance from each of the plurality of bridge portions 144d to one of the pair of first uncoated portion coupling portions 144b adjacent to each other along the extending direction of the edge portion 144a may be substantially equal to a distance from each of the plurality of the bridge portion 144d to the other first uncoated portion coupling portion 144b.

In the case where the first uncoated portion coupling portion 144b and/or the bridge portion 144d is provided in plural as described above, if the distance between the first uncoated portion coupling portions 144b and/or the distance between the bridge portions 144d and/or the distance between the first uncoated portion coupling portion 144b and the bridge portion 144d is uniformly formed, a current flowing from the first uncoated portion coupling portion 144b toward the bridge portion 144d or a current flowing from the bridge portion 144d toward the first uncoated portion coupling portion 144b may be smoothly formed.

The bridge portion 144d may include a notching portion N formed to partially reduce a sectional area of the bridge portion 144d. The sectional area of the notching portion N may be adjusted, for example, by partially reducing the width and/or thickness of the bridge portion 144d. When the notching portion N is provided, electric resistance is increased in the region where the notching portion N is formed, thereby enabling rapid current interruption when overcurrent occurs.

The notching portion N is preferably provided in a region corresponding to the stack number uniform region of the electrode assembly 110 in order to prevent foreign substances generated during rupturing from flowing into the electrode assembly 110. This is because, in this region, the number of overlapping layers of the segments of the uncoated portion 146a is maintained to the maximum and thus the overlapped segments may function as a mask.

The notching portion N may be wrapped with an insulating tape. Then, since the heat generated in the notching portion N is not dissipated to the outside, the notching portion N may be ruptured more quickly when an overcurrent flows through the bridge portion 144d.

FIG. 34 is a top plan view showing the structure of the second current collector 176 according to an embodiment of the present disclosure.

Referring to FIGS. 30 and 34 together, the second current collector 176 is disposed below the electrode assembly 110. In addition, the second current collector 176 may be configured to electrically connect the uncoated portion 146b of the electrode assembly 110 and the battery housing 171. The second current collector 176 is made of a metal material with conductivity and is electrically connected to the bending surface region F of the uncoated portion 146b. In addition, the second current collector 176 is electrically connected to the battery housing 171. The edge portion of the second current collector 176 may be interposed and fixed between the inner surface of the battery housing 171 and the first gasket 178b. Specifically, the edge portion of the second current collector 176 may be interposed between the lower surface of the beading portion 180 of the battery housing 171 and the first gasket 178b. However, the present disclosure is not limited thereto, and the edge portion of the second current collector 176 may be welded to the inner wall surface of the battery housing 171 in a region where the beading portion 180 is not formed.

The second current collector 176 may include a support portion 176a disposed below the electrode assembly 110, a second uncoated portion coupling portion 176b extending from the support portion 176a approximately along the radial direction of the electrode assembly 110 and coupled to the bending surface region F of the uncoated portion 146b, and a housing coupling portion 176c extending from the support portion 176a toward the inner surface of the battery housing 171 approximately along an inclined direction based on the radial direction of the electrode assembly 110 and coupled to the inner surface of the battery housing 171. The second uncoated portion coupling portion 176b and the housing coupling portion 176c are indirectly connected through the support portion 176a, and are not directly connected to each other. Therefore, when an external shock is applied to the cylindrical battery 200 of the present disclosure, it is possible to minimize the possibility of damage to the coupling portion of the second current collector 176 and the electrode assembly 110 and the coupling portion of the second current collector 176 and the battery housing 171. However, the second current collector 176 of the present disclosure is not limited to the structure where

the second uncoated portion coupling portion 176b and the housing coupling portion 176c are only indirectly connected. For example, the second current collector 176 may have a structure that does not include the support portion 176a for indirectly connecting the second uncoated portion coupling portion 176b and the housing coupling portion 176c and/or a structure in which the uncoated portion 146b and the housing coupling portion 176c are directly connected to each other.

The support portion 176a and the second uncoated portion coupling portion 176b are disposed below the electrode assembly 110. The second uncoated portion coupling portion 176b is coupled to the bending surface region F of the uncoated portion 146b. In addition to the second uncoated portion coupling portion 176b, the support portion 176a may also be coupled to the uncoated portion 146b. The second uncoated portion coupling portion 176b and the bending surface region F of the uncoated portion 146b may be coupled by welding. The welding may be replaced with ultrasonic welding or spot welding. The support portion 176a and the second uncoated portion coupling portion 176b are located higher than the beading portion 180 when the beading portion 180 is formed on the battery housing 171.

The support portion 176a has a current collector hole 176d formed at a location corresponding to the hole formed at the core C of the electrode assembly 110. The core C of the electrode assembly 110 and the current collector hole 176d communicating with each other may function as a passage for inserting a welding rod for welding between the terminal 172 and the terminal coupling portion 144c of the first current collector 144 or for irradiating a laser beam.

The current collector hole 176d may have a radius of $0.5 r_c$ or more compared to the radius (r_c) of the hole formed in the core C of the electrode assembly 110. If the radius of the current collector hole 176d is $0.5 r_c$ to $1.0 r_c$, when a vent occurs in the cylindrical battery 200, the phenomenon that the winding structure of the separator or electrodes near the core C of the electrode assembly 110 is pushed out of the core C due to the vent pressure is prevented. When the radius of the current collector hole 176d is larger than $1.0 r_c$, the opening of the core C is maximized, so the electrolyte may be easily injected in the electrolyte injection process.

When the second uncoated portion coupling portion 176b is provided in plural, the plurality of second uncoated portion coupling portions 176b may have a shape extending approximately radially from the support portion 176a of the second current collector 176 toward the sidewall of the battery housing 171. The plurality of second uncoated portion coupling portions 176b may be positioned to be spaced apart from each other along the periphery of the support portion 176a.

The housing coupling portion 176c may be provided in plural. In this case, the plurality of housing coupling portions 176c may have a shape extending approximately radially from the center of the second current collector 176 toward the sidewall of the battery housing 171. Accordingly, the electrical connection between the second current collector 176 and the battery housing 171 may be made at a plurality of points. Since the coupling for electrical connection is made at a plurality of points, the coupling area may be maximized, thereby minimizing electric resistance. The plurality of housing coupling portions 176c may be positioned to be spaced apart from each other along the periphery of the support portion 176a. At least one housing coupling portion 176c may be positioned between the second uncoated portion coupling portions 176b adjacent to each other. The plurality of housing coupling portions 176c

may be coupled to, for example, the beading portion 180 in the inner surface of the battery housing 171. The housing coupling portions 176c may be coupled, particularly, to the lower surface of the beading portion 180 by laser welding. The welding may be replaced with, for example, ultrasonic welding, spot welding, or the like. By coupling the plurality of housing coupling portions 176c on the beading portion 180 by welding in this way, the current path may be distributed radially so that the resistance level of the cylindrical battery 200 is limited to about 4 milliohms or less. In addition, as the lower surface of the beading portion 180 has a shape extending in a direction approximately parallel to the upper surface of the battery housing 171, namely in a direction approximately perpendicular to the sidewall of the battery housing 171, and the housing coupling portion 176c also has a shape extending in the same direction, namely in the radial direction and the circumferential direction, the housing coupling portion 176c may be stably in contact with the beading portion 180. In addition, as the housing coupling portion 176c is stably in contact with the flat portion of the beading portion 180, the two components may be welded smoothly, thereby improving the coupling force between the two components and minimizing the increase in resistance at the coupling portion.

The housing coupling portion 176c may include a contact portion 176e coupled onto the inner surface of the battery housing 171 and a connection portion 176f for connecting the support portion 176a and the contact portion 176e.

The contact portion 176e is coupled onto the inner surface of the battery housing 171. In the case where the beading portion 180 is formed on the battery housing 171, the contact portion 176e may be coupled onto the beading portion 180 as described above. More specifically, the contact portion 176e may be electrically coupled to the flat portion formed at the lower surface of the beading portion 180 formed on the battery housing 171, and may be interposed between the lower surface of the beading portion 180 and the first gasket 178b. In this case, for stable contact and coupling, the contact portion 176e may have a shape extending on the beading portion 180 by a predetermined length along the circumferential direction of the battery housing 171.

The connection portion 176f may be bent at an obtuse angle. The bending point may be higher than the middle point of the connection portion 176f. When the connection portion 176f is bent, the contact portion 176e may be stably supported on the flat surface of the beading portion 180. The connection portion 176f is divided into a lower portion and an upper portion based on the bending point, and the lower portion may have a greater length than the upper portion. In addition, the lower portion of the bending point may have a greater inclination angle based on the surface of the support portion 176a than the upper portion. When the connection portion 176f is bent, a pressure (force) applied in the vertical direction of the battery housing 171 may be buffered. For example, in the process of sizing the battery housing 171, when a pressure is transmitted to the contact portion 176e so that the contact portion 176e moves vertically toward the support portion 176b, the bending point of the connection portion 176f moves upward, so that the shape of the connection portion 176 is deformed to buffer the stress.

Meanwhile, the maximum distance from the center of the second current collector 176 to the end of the second uncoated portion coupling portion 176b along the radial direction of the electrode assembly 110 is preferably equal to or smaller than the inner diameter of the battery housing 171 in a region where the beading portion 180 is formed, namely the minimum inner diameter of the battery housing

171. This is to prevent the end of the second uncoated portion coupling portion 176b from pressing the edge of electrode assembly 110 during the sizing process of compressing the battery housing 171 along the height direction.

The second uncoated portion coupling portion 176b includes a hole 176g. The hole 176g may be used as a passage through which the electrolyte may move. The welding pattern 176h formed by welding between the second uncoated portion coupling portion 176b and the bending surface region F may have a structure to extend along the radial direction of the electrode assembly 110. The welding pattern 176h may be a line pattern or a dot array pattern.

The welding pattern 176h corresponds to the welding region. Therefore, it is preferable that the welding pattern 176h overlaps by 50% or more with the stack number uniform region b1 of the bending surface region F located in the lower portion of the electrode assembly 110. The welding pattern 176h that does not overlap with the stack number uniform region b1 may overlap with the stack number decrease region b2. More preferably, the entire welding pattern 176h may overlap with the stack number uniform region b1 of the bending surface region F. In the bending surface region F at the upper portion of the point where the welding pattern 176h is formed, the stack number uniform region b1 and, optionally, the stack number decrease region b2 preferably have the stack number of 10 or more.

The outer diameters of the first current collector 144 and the second current collector 176 described above are different from each other. The outer diameter is an outer diameter of the outer edge of the contact area between the bending surface region F and the current collector. The outer diameter is defined as a maximum value of the distance between two points where a straight line passing through the center of the core C of the electrode assembly meets the edge of the contact area. Since the second current collector 176 is located inside the beading portion 180, its outer diameter is smaller than that of the first current collector 144. In addition, the length of the welding pattern 144f of the first current collector 144 is longer than the length of the welding pattern 176h of the second current collector 176. Preferably, the welding pattern 144f and the welding pattern 176h may extend toward the outer circumference from substantially the same point based on the center of the core C.

The welding region of the first current collector 144 and the second current collector 176 may have a radial structure based on the core center of the electrode assembly.

The welding structure of the upper bending surface region F and the first current collector 144 and/or the welding structure of the lower bending surface region F and the second current collector 176 work synergistically with the winding structure of the electrode assembly to prevent or alleviate the collapse of the core. As a result, it is possible to suppress the electrode from rotating while the cylindrical battery is repeatedly charged and discharged, for example, while the cylindrical battery is charged and discharged beyond the effective usage cycle.

The cylindrical battery 170, 180, 200, 220 according to an embodiment of the present disclosure have an advantage in that electrical connection can be performed at the upper portion thereof.

FIG. 35 is a top plan view illustrating a state in which a plurality of cylindrical batteries 200 are electrically connected, and FIG. 36 is a partially enlarged view of FIG. 35. The cylindrical battery 200 may be replaced with a cylindrical battery 170, 180, 220 having a different structure.

Referring to FIGS. 35 and 36, a plurality of cylindrical batteries 200 may be connected in series and in parallel at an

upper portion of the cylindrical batteries 200 using a bus bar 210. The number of cylindrical batteries 200 may be increased or decreased in consideration of the capacity of the battery pack.

In each cylindrical battery 200, the terminal 172 may have a positive polarity, and the flat surface 171a around the terminal 172 of the battery housing 171 may have a negative polarity, or vice versa.

Preferably, the plurality of cylindrical batteries 200 may be arranged in a plurality of columns and rows. Columns are provided in a vertical direction on the drawing, and rows are provided in a left and right direction on the drawing. In addition, in order to maximize space efficiency, the cylindrical batteries 200 may be arranged in a closest packing structure. The closest packing structure is formed when an equilateral triangle is formed by connecting the centers of the terminals 172 exposed out of the battery housing 171 to each other. Preferably, the bus bar 210 connects the cylindrical batteries 200 arranged in the same column in parallel to each other, and connects the cylindrical batteries 200 arranged in two neighboring columns in series with each other.

Preferably, the bus bar 210 may include a body portion 211, a plurality of first bus bar terminals 212 and a plurality of second bus bar terminals 213 for serial and parallel connection.

The body portion 211 may extend along the column of the cylindrical batteries 200 between neighboring terminals 172. Alternatively, the body portion 211 may extend along the column of the cylindrical batteries 200, and the body portion 211 may be regularly bent like a zigzag shape.

The plurality of first bus bar terminals 212 may extend in one side direction of the body portion 211 and may be electrically coupled to the terminal 172 of the cylindrical battery 200 located in one side direction. The electrical connection between the first bus bar terminal 212 and the terminal 172 may be achieved by laser welding, ultrasonic welding, or the like.

The plurality of second bus bar terminals 213 may extend in the other side direction of the body portion 211 and may be electrically coupled to the flat surface 171a around the terminal 172 located in the other side direction. The electrical coupling between the second bus bar terminal 213 and the flat surface 171a may be performed by laser welding, ultrasonic welding, or the like.

Preferably, the body portion 211, the plurality of first bus bar terminals 212 and the plurality of second bus bar terminals 213 may be made of one conductive metal plate. The metal plate may be, for example, an aluminum plate or a copper plate, but the present disclosure is not limited thereto. In a modified example, the body portion 211, the plurality of first bus bar terminals 212 and the second bus bar terminals 213 may be manufactured as separate pieces and then coupled to each other by welding or the like.

The cylindrical battery 200 of the present disclosure as described above has a structure in which resistance is minimized by enlarging the welding region by means of the bending surface region F, multiplexing current paths by means of the second current collector 176, minimizing a current path length, or the like. The AC resistance of the cylindrical battery 200 measured through a resistance meter between the positive electrode and the negative electrode, namely between the terminal 172 and the flat surface 171a around the terminal 172, may be approximately 0.5 milliohms to 4 milliohms, preferably 1 milliohms to 4 milliohms, suitable for fast charging.

In the cylindrical battery 200 according to the present disclosure, since the terminal 172 having a positive polarity and the flat surface 171a having a negative polarity are located in the same direction, it is easy to electrically connect the cylindrical batteries 200 using the bus bar 210.

In addition, since the terminal 172 of the cylindrical battery 200 and the flat surface 171a around the terminal 172 have a large area, the coupling area of the bus bar 210 may be sufficiently secured to sufficiently reduce the resistance of the battery pack including the cylindrical battery 200.

In addition, since electrical wiring may be performed on the upper portion of the cylindrical battery 200, there is an advantage in maximizing the energy density per unit volume of the battery module/pack.

The cylindrical battery according to the above embodiments (modifications) may be used to manufacture a battery pack.

FIG. 37 is a diagram schematically showing a battery pack according to an embodiment of the present disclosure.

Referring to FIG. 37, a battery pack 300 according to an embodiment of the present disclosure includes an aggregate in which cylindrical batteries 301 are electrically connected, and a pack housing 302 for accommodating the aggregate. The cylindrical battery 301 may be any one of the batteries according to the above embodiments (modifications). In the drawing, components such as a bus bar, a cooling unit, and an external terminal for electrical connection of the cylindrical batteries 301 are not depicted for convenience of illustration.

The battery pack 300 may be mounted to a vehicle. The vehicle may be, for example, an electric vehicle, a hybrid electric vehicle, or a plug-in hybrid vehicle. The vehicle includes a four-wheeled vehicle or a two-wheeled vehicle.

FIG. 38 is a diagram schematically showing a vehicle including the battery pack 300 of FIG. 37.

Referring to FIG. 38, a vehicle V according to an embodiment of the present disclosure includes the battery pack 300 according to an embodiment of the present disclosure. The vehicle V operates by receiving power from the battery pack 300 according to an embodiment of the present disclosure.

According to an aspect of the present disclosure, it is possible to prevent the core from collapsing by maintaining the symmetry and circularity of the electrode assembly even if swelling phenomenon occurs by adjusting relative positions of a positive electrode end and a negative electrode end at the core and the outer circumference of the electrode assembly of the cylindrical battery.

According to another aspect of the present disclosure, it is possible to provide a cylindrical battery including an electrode assembly with a structure capable of improving the collapse phenomenon of the core.

According to still another aspect of the present disclosure, the internal resistance of the battery may be reduced and the energy density may be increased by using the uncoated portion itself protruding on the upper and lower portions of the electrode assembly as an electrode tab.

According to another aspect of the present disclosure, by improving the structure of the uncoated portion of the electrode assembly so that the electrode assembly and the inner circumference of the battery housing do not interfere in the process of forming the beading portion of the battery housing, it is possible to prevent a short circuit from occurring inside the cylindrical battery due to partial deformation of the electrode assembly.

According to still another aspect of the present disclosure, by improving the structure of the uncoated portion of the electrode assembly, it is possible to prevent the uncoated

portion from being torn when the uncoated portion is bent, and it is possible to improve the welding strength of the current collector by sufficiently increasing the number of overlapping layers of the uncoated portion.

According to still another aspect of the present disclosure, by applying a segment structure to the uncoated portion of the electrode and optimizing the dimensions (width, height, separation pitch) of the segments to sufficiently increase the segment stack number of the area used as the welding target area, it is possible to improve the properties of the area where the current collector is welded.

According to still another aspect of the present disclosure, an electrode assembly having improved energy density and reduced resistance may be provided by applying a structure in which a current collector is welded to a broad area of the bending surface region formed by bending the segments.

According to still another aspect of the present disclosure, a cylindrical battery having an improved design so that electrical wiring can be performed at the upper portion thereof may be provided.

According to still another aspect of the present disclosure, by improving the structure of the uncoated portion adjacent to the core of the electrode assembly, the cavity in the core of the electrode assembly is prevented from being blocked when the uncoated portion is bent, so that the electrolyte injection process and the process of welding the battery housing (or, terminal) and the current collector may be easily performed.

According to still another aspect of the present disclosure, it is possible to provide a cylindrical battery having a structure in which the internal resistance is low, an internal short circuit is prevented, and the welding strength between the current collector and the uncoated portion is improved, and a battery pack and a vehicle including the cylindrical battery.

In particular, the present disclosure may provide a cylindrical battery having a ratio of diameter to height of 0.4 or more and a resistance of 4 milliohm or less, and a battery pack and a vehicle including the cylindrical battery.

The present disclosure has been described in detail. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the scope of the disclosure will become apparent to those skilled in the art from this detailed description.

What is claimed is:

1. A cylindrical battery, comprising:

an electrode assembly in which a first electrode, a second electrode, and a separator interposed therebetween are wound around a winding axis of the electrode assembly, so as to define a hollow core and an outer circumference of the electrode assembly; and

a cylindrical battery housing accommodating the electrode assembly,

wherein a cross-section of the electrode assembly perpendicular to a winding axis direction includes a first fan-shaped region and a second fan-shaped region, wherein the first fan-shaped region is defined by a first straight line and a second straight line passing from a center of the core through a core side end of the first electrode and a core side end of the second electrode, respectively, and an outer circumference of the first fan-shaped region is defined as a stress vulnerable region, and

wherein the second fan-shaped region is defined by a third straight line and a fourth straight line passing from the

center of the core through an outer circumference side end of the first electrode and an outer circumference side end of the second electrode, respectively, and an outer circumference of the second fan-shaped region is defined as a stress amplification region, and
 5 wherein the electrode assembly has a winding structure in which at least the outer circumference side end of the first electrode of the stress amplification region is spaced apart from an inside of the stress vulnerable region along a circumferential direction of the cross-
 10 section of the electrode assembly,
 wherein in the winding structure, a circumferential angle of the stress vulnerable region has an angle of 30 degrees or more and 180 degrees or less, and a circumferential angle of the stress amplification region has an
 15 angle in a range of 1-degrees to 90 degrees.

2. The cylindrical battery according to claim 1, wherein the battery has a ratio of a diameter to a height greater than 0.4.

3. The cylindrical battery according to claim 2,
 20 wherein the cylindrical battery has a form factor of 46110, 4875, 48110, 4880, 4680, or 4695.

4. A battery pack, comprising a plurality of cylindrical batteries according to claim 1.

5. A vehicle, comprising the battery pack according to
 25 claim 4.

6. The cylindrical battery according to claim 1, wherein a diameter of the cross-section is greater than 40 mm.

7. The cylindrical battery according to claim 1,
 30 wherein a region of the outer circumference of the electrode assembly including the outer circumference side end of the first electrode contacts an inner surface of the battery housing.

8. The cylindrical battery according to claim 1,
 35 wherein in the winding structure, the stress amplification region is spaced apart from the stress vulnerable region along the circumferential direction of the cross-section of the electrode assembly.

9. The cylindrical battery according to claim 1,
 40 wherein the cross-section includes a first semicircular region and a second semicircular region based on a diametric line segment,
 wherein the diametric line segment passes through a center of the core and is perpendicular to a straight line dividing a circumferential angle of the stress vulnerable
 45 region into equal angles,
 wherein in the winding structure, the stress vulnerable region is located in the first semicircular region and the stress amplification region is located in the second semicircular region.

10. The cylindrical battery according to claim 1,
 50 wherein in the winding structure at least a part of the stress amplification region overlaps with a third fan-shaped region in a second semicircular region, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core.

11. The cylindrical battery according to claim 10,
 wherein in the winding structure at least a part of the stress amplification region overlaps with a fifth straight
 60 line that divides a circumferential angle of the third fan-shaped region into equal angles.

12. The cylindrical battery according to claim 1,
 65 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the third straight line and the fourth straight line are located between the second straight line and the fifth straight line based on the circumferential direction of the cross-section.

13. The cylindrical battery according to claim 1,
 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
 wherein in the winding structure the third straight line and the fourth straight line are located between the first straight line and the fifth straight line based on the circumferential direction on the cross-section.

14. The cylindrical battery according to claim 1,
 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
 wherein in the winding structure the fourth straight line is located between the fifth straight line and the first straight line, and the third straight line is located between the fourth straight line and the second straight line based on the circumferential direction on the cross-section.

15. The cylindrical battery according to claim 1,
 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
 wherein in the winding structure the fourth straight line is located between the fifth straight line and the first straight line, and the third straight line is located between the fifth straight line and the second straight line based on the circumferential direction on the cross-section.

16. The cylindrical battery according to claim 1,
 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
 wherein in the winding structure the fourth straight line is located between the fifth straight line and the second straight line, and the third straight line is located between the fourth straight line and the first straight line based on the circumferential direction on the cross-section.

17. The cylindrical battery according to claim 1,
 wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
 wherein in the winding structure the fourth straight line is located between the fifth straight line and the second straight line, and the third straight line is located between the fifth straight line and the first straight line based on the circumferential direction on the cross-section.

18. The cylindrical battery according to claim 1,
 wherein the winding structure has a circumferential angle between the first straight line and the fourth straight line that is greater than a circumferential angle between the second straight line and the third straight line, based on the circumferential direction on the cross-section.

19. The cylindrical battery according to claim 1,
 wherein the winding structure has a circumferential angle between the second straight line and the fourth straight line that is greater than a circumferential angle between

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the first straight line and the third straight line, based on the circumferential direction on the cross-section.

20. The cylindrical battery according to claim 1, wherein the winding structure has a circumferential angle of the stress amplification region that is smaller than a circumferential angle of the stress vulnerable region, based on the center of the core.

21. The cylindrical battery according to claim 1, wherein the winding structure of the electrode assembly is maintained during charging and discharging in a range of 200 cycles to 900 cycles.

22. The cylindrical battery according to claim 1, wherein the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, wherein the first electrode includes a plurality of positive electrode winding turns and the second electrode includes a plurality of negative electrode winding turns, wherein at least a part of the plurality of negative electrode winding turns is provided adjacent to a core side of the electrode assembly, and wherein at least a part of the plurality of negative electrode winding turns extends in a radial direction.

23. The cylindrical battery according to claim 22, wherein the separator includes a plurality of winding turns provided at an inner side of the plurality of negative electrode winding turns.

24. The cylindrical battery according to claim 1, wherein the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, wherein the core side end of the second electrode extends further in a direction opposite to the winding-axis direction than the core side end of the first electrode, so as to define at least a part of an innermost winding turn, and wherein the outer circumference side end of the second electrode extends further in the winding direction than the outer circumference side end of the first electrode, so as to define at least a part of an outermost winding turn.

25. The cylindrical battery according to claim 1, wherein in the winding structure, the circumferential angle of the stress vulnerable region is has an angle in a range of 87 degrees or more, and the circumferential angle of the stress amplification region has an angle in a range of 32 degrees or less.

26. The cylindrical battery according to claim 1, wherein the battery housing includes an open end and a bottom portion opposing thereto and accommodates the electrode assembly in a space between the open end and the bottom portion, and wherein the battery housing is electrically connected to one of the first electrode or the second electrode, so as to have a first polarity.

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27. The cylindrical battery according to claim 26, further comprising:
a sealing body sealing the open end of the battery housing;
and
a terminal electrically connected to the other of the first electrode or the second electrode, so as to have a second polarity,
wherein the terminal has a surface exposed to an exterior of the cylindrical battery.

28. The cylindrical battery according to claim 1, wherein the first electrode includes a first uncoated portion at a long side end and extending along the winding axis direction of the electrode assembly, wherein the first uncoated portion extends and protrudes outward from the separator through one end of the electrode assembly and is in a bent state extending in a radial direction of the electrode assembly to define a first bending surface region, and wherein the cylindrical battery further comprises a first current collector coupled to the first bending surface region.

29. The cylindrical battery according to claim 28, wherein the first uncoated portion includes a plurality of segments along the winding-axis direction of the electrode assembly, and wherein the plurality of segments are in a bent state extending in the radial direction of the electrode assembly to define the first bending surface region.

30. The cylindrical battery according to claim 28, wherein the second electrode includes a second uncoated portion at a long side end and extending along the winding axis direction, wherein the second uncoated portion extends and protrudes outward from the separator through the other end of the electrode assembly and is in a bent state extending toward the core to define a second bending surface region, and wherein the cylindrical battery further comprises a second current collector coupled to the second bending surface region.

31. The cylindrical battery according to claim 30, wherein the second uncoated portion includes a plurality of segments along the winding axis direction of the electrode assembly, and wherein the plurality of segments are in a bent state extending in the radial direction of the electrode assembly to define the second bending surface region.

32. The cylindrical battery according to claim 28, wherein a coupling region of the first current collector has a radial structure.

33. The cylindrical battery according to claim 30, wherein the welding a coupling region of the second current collector has a radial structure.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

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DATED : September 9, 2025
INVENTOR(S) : Yong-Hwan Kwon et al.

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 105, Line 16, Claim 1, "1-degrees" should read --10 degrees--

Signed and Sealed this
Twenty-fourth Day of February, 2026



John A. Squires
Director of the United States Patent and Trademark Office

